# Orchard Pollination Planting Grid and West Windbreak

Prairie Plants — inventory of October 4, 2026; planting status updated October 18, 2026 — orchard plot 237 × 600 ft plus a 35 ft west windbreak — 13 positions per row — 18 ft in-row (E–W) × 20 ft rows (N–S), triangular offset

Sources: Inventory_PrairiePlants.xlsx (what was found at Prairie, saved October 4, 2026), matched to Trees.xlsx, Fruit sheet (cultivar, rootstock, fertility, and flowering-group data, including the September 24 bloom research). Owner-reported losses, outside-orchard plans, and rootstock planting are recorded separately; only living trees assigned to the orchard are counted in the plan. Canonical JSON records now drive this page; the linked workbooks are retained as original source records. See the property data guide.

R4-C9: Tandgy green Urban Columnar (tag AL-UrbGrn, inventory row 64) not planted in the orchard. This living tree is planted outside the orchard; SiteRef D13-E0-S20. Leave this spot open; the plan is a new Haralson (apple) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock M.7, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan.
R5-C9: Tasty Red Urban Columnar (tag AL-UrbR, inventory row 65) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E46-S20. Leave this spot open; the plan is a new Williams' Pride (apple) in 2027. It is ordered and has not arrived: Cummins Nursery order 039970, 2026-10-27, ships to Idaho. Rootstock B.118, grade 1, Spring 2027. Described in the planting plan.
R5-C10: Blushing Delight Urban Columnar (tag AL-UrbGld, inventory row 63) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E42-S20. Leave this spot open; the plan is a new Priscilla (apple) in 2027. Described in the planting plan.
R8-C1: Pineapple (tag PR-Pineap, inventory row 108) reported dead October 11, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Moonglow, stand-in for Harrow Delight (pear) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock OHxF 97, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan.
R9-C2: Royalton Sweet (tag CR-Royalton, inventory row 80) reported dead October 17, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Black Pearl (cherry) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order XFEEV5ANY, 2026-10-28, ships to Idaho. Rootstock Mazzard, grade Hand Select, Spring 2027. Described in the planting plan.
R11-C10: JH Hale (tag PC-Hale, inventory row 88) reported dead October 17, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Intrepid (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Seedling, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan.
R11-C12: Canadian Harmony (tag PC-Can Harm, inventory row 84) reported dead October 18, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Contender (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Lovell, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan.
R12-C11: BlazingStar (tag PC-Blaz Star, inventory row 83) reported dead October 18, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Redhaven, stand-in for Harrow Diamond (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Seedling, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan.
M-111 apple rootstock planted at R6-C3. No scion cultivar is recorded; not counted as a fruit tree or pollen partner until grafting and cultivar identification are recorded.
All 106 living orchard fruit trees are planted (October 2026), per the owner's update of October 18, 2026.

| ORCHARD PLOT | WEST WINDBREAK | TRACTOR ALLEY | TREE SPACING | ROW SPACING | DIAGONAL NEIGHBOR | ORCHARD TREES | OPEN POSITIONS | ROWS | WINDBREAK · HEDGE · GUILDS |
|---|---|---|---|---|---|---|---|---|---|
| 237 × 600 ft | 35 ft, 2 rows | 7 ft clear | 18 ft in-row | 20 ft N–S | 21.9 ft | 107 | 75 | 15 (S→N) | 59 + 19 + 40 |

Width, west to east: the 35 ft windbreak strip (2 plant rows with a 10 ft walking lane between them, then the tractor alley), then the 237 ft orchard plot. C1 trunks stand 3 ft inside the orchard's west line, 38 ft from the fence. 13 staggered positions span 225 ft, leaving 9 ft to the east line beside the offset rows' C13 trees (18 ft beside the others); a 14th position would need 243 ft. East of the orchard: 28 ft of open ground for turning at the row ends, the 22 ft driveway, then 10 ft to the east fence (332 ft from the west fence), with the elderberry hedge 5 ft inside it. Depth: 15 rows span 280 ft between the outer row centers. With the windbreak ends, the planted block uses about 300 ft of the 600 ft depth, leaving about 300 ft for headlands and future rows. Distances are tree centers, before mature-canopy clearance; the 18 × 20 ft spacing is a planning assumption, not a mature-spacing recommendation for every rootstock.

## Planting Map — South at Top, West Fence on the Right

South is up, the west fence and windbreak are on the right, and east is on the left: the view from the north end looking south. Rows start at the west alley (C1, on the right) and fill toward the east (left), so open spots are at the east ends, ready to fill in from the east later, except that in R11–R12 the apricots are right-justified, plums are placed at C6 in R11 and C5–C6 in R12, and peaches are left-justified, so the open spots there are between the blocks, and at R4-C9, R5-C9, R5-C10, R8-C1, R9-C2, R11-C10, R11-C12, R12-C11 after the owner's planting updates. Rows from south to north (top to bottom): apples in 5 rows (13 + 13 + 12 + 11 + 3) plus reserved R1 (13 spots); pears in 2 rows (13 + 11); cherries in 2 rows (7 + 8); apricots at the west end (3 + 2), plums at C6 in R11 and C5–C6 in R12 (1 + 2), and peaches at the east end (4 + 4) share R11–R12; R13–R15 are open for trees you buy or graft later. R1 is held for 13 grafted rootstock apples. The C1 trees in R2–R12 have berry guilds on their alley side (green dots); exact spots are on the site plan. The windbreak band is schematic; exact positions are in the windbreak table and site plan, which also shows the elderberry hedge along the east fence. Full names, tags, and notes are listed by position in the inventory.

▲ SOUTH — the closest juniper or red cedar is 200+ ft beyond this end

| | ← EAST 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Apple — 13 reserved for rootstock apples | PXC graft | PXC graft | NVM graft | NVM graft | SNS graft | SNS graft | AKN graft | AKN graft | RDF graft | RDF graft | RDF graft | CNC graft | CNC graft | R1 |
| Apple — 13 trees; 0 open | ✓ FRQ FG 4* | ✓ FC FG 4 | ✓ CO FG 3 S | ✓ AB FG 3 ▲ | ✓ GL FG 4 S | ✓ JG FG 4 ▲ | ✓ PL FG 3 | ✓ CL FG 3 | ✓ HC FG 4 | ✓ FJ FG 4 | ✓ JG FG 4 ▲ | ✓ BB FG 4 S | ✓ FOX FG 3 ◔ | R2 |
| Apple — 13 trees; 0 open | ✓ NP FG 3 S | ✓ UNK FG ? | ✓ RD FG 4 | ✓ GO FG 4 | ✓ FA FG 4 | ✓ HC FG 4 | ✓ UNK FG ? | ✓ FA FG 4 | ✓ PL FG 3 | ✓ MC FG 3 | ✓ KB FG 4 | ✓ CL FG 3 | ✓ KER FG 2? ◔ | R3 |
| Apple — 12 trees; 1 open | ✓ DLG FG 2 | ✓ HR FG 3* | ✓ AF FG 4 | ✓ UNK FG ? | HAL ord UG outside | ✓ UNK FG ? | ✓ WC FG 2 | ✓ ZT FG 1 | ✓ UNK FG ? | ✓ GD FG 4 | ✓ HC FG 4 | ✓ UNK FG ? | ✓ CO FG 3 S ◔ | R4 |
| Apple — 11 trees; 2 open | ✓ KER FG 2? | ✓ CC FG 3 | ✓ HR FG 3* | Open UD outside PSC 2027 | WLP ord UR outside | ✓ ZT FG 1 | ✓ MT FG 2 | ✓ CL FG 3 | ✓ SS FG 3 | ✓ LB FG 2 | ✓ HG FG 3 | ✓ ID FG 2 | ✓ CAM FG 3 ◔ | R5 |
| Apple — 3 trees; 10 open | FRD ord | PSC 2027 | DAY 2027 | HAL ord | PTN ord | WLP ord | WLP ord | FRD ord | FRD ord | DAY 2027 | ✓ RS M-111 | ✓ GS FG 3 S | ✓ KER FG 2? ◔ | R6 |
| Pear — 13 trees; 0 open | ✓ BUT ? | ✓ KF ? | ✓ CE Self | ✓ SMC ? | ✓ PKR ? | ✓ SMC ? | ✓ BNT ? | ✓ CE Self | ✓ AU ? | ✓ RB ? | ✓ OG Asian | ✓ SA Asian | ✓ YHC ? ◔ | R7 |
| Pear — 11 trees; 2 open | SKL ord | ✓ CE Self | ✓ BT ? | ✓ AU ? | ✓ SD ? | ✓ HHC ? ◥ | ✓ AU ? | ✓ MW ? | ✓ BT ? | ✓ HHC ? ◥ | ✓ HG Asian | ✓ HA Asian | HDL ord PF dead ◔ | R8 |
| Cherry — 7 trees; 6 open | SUR ord | SUR ord | BGY ord | BPL ord | BGY ord | ✓ NKI ? | ✓ HG ? | ✓ SWT ? | ✓ NKI ? | ✓ ST Self | ✓ BB Period 3 | BPL ord RS dead | ✓ BB Period 3 ◔ | R9 |
| Cherry — 8 trees; 5 open | MON ord | MON ord | SUR ord | BTR ord | SUR ord | ✓ MNT Self | ✓ SWT ? | ✓ NKJ ? | ✓ MNT Self | ✓ SWT ? | ✓ BT Period 1 | ✓ LP Self | ✓ BT Period 1 ◔ | R10 |
| Peach · plum · apricot — 8 trees; 5 open | ✓ RC Self | CR ord CH dead | ✓ FT Self | INT ord JH dead | ✓ RH Self | ✓ CR Self | INT ord | ✓ IPL Self | HT ord | BNG ord | ✓ GT Self | ✓ PN Self | ✓ MP Self ◔ | R11 |
| Peach · plum · apricot — 8 trees; 5 open | ✓ EA Self | ✓ UNK ? | HRD ord BS dead | ✓ UNK ? | ✓ RC Self | RC ord | INT ord | ✓ IPL Self | ✓ IPL Self | DMS ord | STN ord | ✓ MP Self | ✓ HT Self ◔ | R12 |
| Future trees — 13 open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | R13 |
| Future trees — 13 open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | R14 |
| Future trees — 13 open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | — Open | R15 |

Tractor alley · 7 ft clear · Hazelnut × 34 · 20 ft from fence · every 9 ft · Walking lane · 10 ft · Red mulberry × 25 · 5 ft from fence · every 12.5 ft · WEST FENCE

▼ NORTH

Apple FG 1 · Apple FG 2 · Apple FG 3 · Apple FG 4 · Apple — relative bloom only; FG unverified · Apple — unidentified or bloom unconfirmed · Planted apple rootstock (RS) — no scion recorded · Pear — European · Pear — Asian · Pear — Perry · Cherry · Apricot · Peach · Plum · Dead tree: position open for a replacement · Not planted here: tree planned outside; replacement in 2027 · Planned 2027: recommended cultivar for an open spot (a dashed border marks a spot held for a dead or relocated tree; "graft" marks a scion for one of your rootstocks) · Ordered 2027: bought, not yet arrived (blue cell, marked ord). Not a pollen partner until it is in the ground · Windbreak: Red mulberry · Windbreak: Hazelnut · Berry guild on the alley side of the C1 tree

▲ Dashed border = **triploid**, excluded as a pollen donor and placed near two different diploid cultivars. ★ Gold glow = **good pollen donor in the workbook**, not a guarantee of compatibility with every nearby tree.

◥ Yellow corner = **Pseudomonas (bacterial canker) positive, untreated** in the inventory. When one tag covers several trees and only some tested positive, the inventory does not say which, so all of them are flagged.

✓ = planted. Every living orchard fruit tree is planted (October 2026), and so is the M-111 rootstock at R6-C3.

Green cells are the recommended 2027 plantings for every open spot in R1 to R12 (26 apples, 2 pears, 6 peaches, 12 cherries, 2 plums, 1 apricot), shown by cultivar code with the year; the 13 marked graft are scions for the rootstocks you own. 32 of those spots are ordered and show in blue, marked ord; they have not arrived. The rootstocks, cautions, and buy list are in the 2027 planting plan. Planned and ordered trees are not counted as pollen partners for the trees on this page.

**S** = self-fertile according to the planning notes; **?** = information missing or needing confirmation; **UNK** / **SWT** = unidentified tree (not counted as a pollen donor until identified; see the possibility lists for the apples and for the cherries and peaches). Hover over a tree for its cultivar profile, or click or tap it to jump to its row in the inventory table, which links to the cultivar guide. Different trees of the same cultivar, including color sports, are not counted as different pollen partners.

Apple **FG** means flowering group, not harvest season. ***** after a group number means sources differ; the planning value follows Orange Pippin. **FG 2?** marks a provisional estimate. **Midseason; FG ?** records nursery bloom guidance without inventing a numbered group.

## Ordered, not yet arrived

These trees are bought for the blue cells. A stand-in is the tree that was ordered for that planned cultivar. None of them is in the ground, so none is counted as a pollen partner.

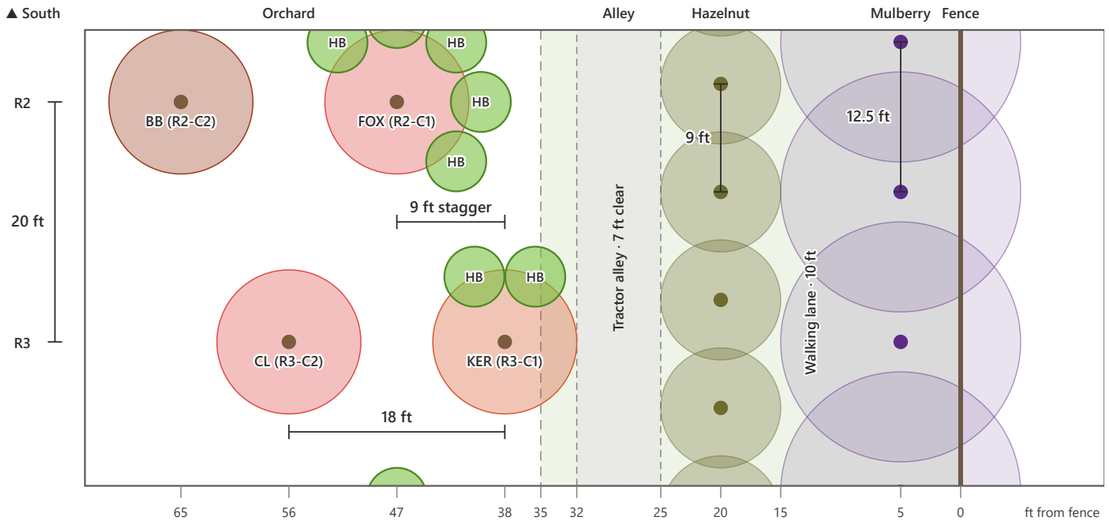
| Cultivar | Spots | Rootstock and grade | Nursery |
|---|---|---|---|
| **Williams' Pride** | 3: R6-C7, R6-C8, R5-C9 | B.118, grade 1 | Cummins Nursery 039970, 2026-10-27 |
| **Freedom** | 3: R6-C5, R6-C6, R6-C13 | M.111, grade Hand Select | Roots to Fruits Nursery RTF12645, 2026-10-27<br>Payment still due |
| **Pristine** | 1: R6-C9 | B.118, grade Hand Select | Roots to Fruits Nursery RTF12645, 2026-10-27<br>Payment still due |
| **Haralson** | 2: R4-C9, R6-C10 | M.7, grade Hand Select | Roots to Fruits Nursery RTF12645, 2026-10-27<br>Payment still due |
| **Intrepid** | 1: R11-C10 | Seedling, grade Field Ready | Roots to Fruits Nursery RTF12645, 2026-10-27<br>Payment still due |
| **Contender** | 1: R11-C12 | Lovell, grade Field Ready | Roots to Fruits Nursery RTF12645, 2026-10-27<br>Payment still due |
| **Reliance** | 1: R12-C8 | Seedling, grade Field Ready | Roots to Fruits Nursery RTF12645, 2026-10-27<br>Payment still due |
| **Redhaven**<br>Stand-in for planned Harrow Diamond | 1: R12-C11 | Seedling, grade Field Ready | Roots to Fruits Nursery RTF12645, 2026-10-27<br>Payment still due |
| **Moonglow**<br>Stand-in for planned Harrow Delight | 1: R8-C1 | OHxF 97, grade Hand Select | Roots to Fruits Nursery RTF12645, 2026-10-27<br>Payment still due |
| **Seckel** | 1: R8-C13 | OHxF 87, grade 4-5 ft whip | Raintree Nursery U2TM4QUWD, 2026-10-27 |
| **Harcot** | 1: R11-C5 | Mariana, grade Feathered | Trees of Antiquity 2R8PX8Q8P, 2026-10-27 |
| **Stanley** | 1: R12-C3 | Mariana, grade Feathered | Trees of Antiquity 2R8PX8Q8P, 2026-10-27 |
| **Damson** | 1: R12-C4 | Mariana, grade Feathered | Trees of Antiquity 2R8PX8Q8P, 2026-10-27 |
| **Intrepid** | 2: R11-C7, R12-C7 | Seedling, grade Field Ready | Roots to Fruits Nursery WZTA0LZIJ, 2026-10-28 |
| **Black Pearl** | 2: R9-C2, R9-C10 | Mazzard, grade Hand Select | Roots to Fruits Nursery XFEEV5ANY, 2026-10-28 |
| **Burgundy Pearl** | 2: R9-C9, R9-C11 | Mazzard, grade Hand Select | Roots to Fruits Nursery XFEEV5ANY, 2026-10-28 |
| **Surefire** | 4: R10-C9, R10-C11, R9-C12, R9-C13 | Mahaleb, grade 1-year | Orange Pippin Trees USA US332070, 2026-10-28<br>Payment still due |
| **Montmorency** | 2: R10-C12, R10-C13 | Not stated, grade EZ-Pick fruiting size | Willis Orchard Company 1101164703766, 2026-10-28 |
| **Bing** | 1: R11-C4 | Not stated, grade 6-7 ft | Willis Orchard Company 1101164703766, 2026-10-28 |
| **Black Tartarian** | 1: R10-C10 | Not stated, grade 6-7 ft branched | Willis Orchard Company 1101164703766, 2026-10-28 |

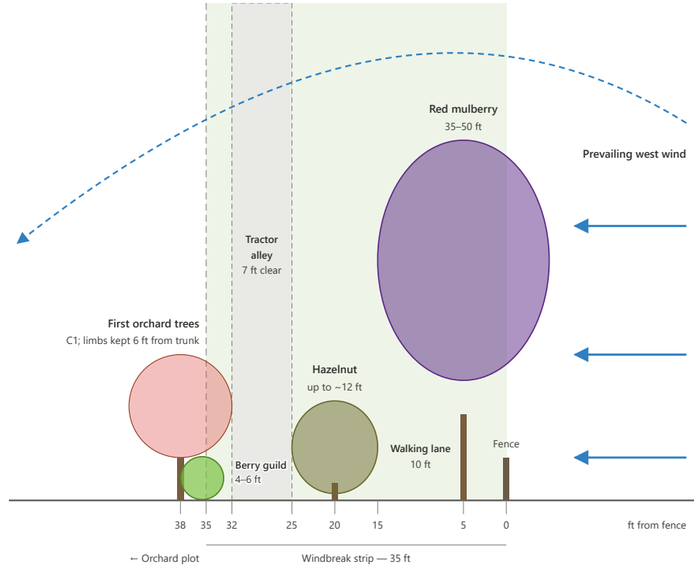
## West Windbreak (35 ft Strip) and East Elderberry Hedge

Two north–south rows step down from the west fence toward the orchard: red mulberry (tallest) along the fence and hybrid hazelnut 15 ft east of it, so the prevailing west wind is pushed up and over the fruit trees. A 10 ft walking lane under the mulberries leads to the hazelnuts for picking. The strip is added west of the 237 ft orchard plot; the orchard was not reduced. Both rows spread the plants you have along the orchard and reach at least 6 ft past R1 and R15. The 41 berry bushes came out of the strip to make room for the lane and now form guilds on the alley side of the C1 orchard trees in R2–R12. The elderberry hedge runs 5 ft inside the east fence beside the driveway at its proper spacing from the south end, with 19 positions marked for plants to add later. East–west positions are measured from the west fence and north–south positions from the R1 row line. General notes are beside the site plan.

**Blow-up plan view, to scale: a 38 ft stretch of the windbreak beside the first apple rows (south up, west on the right)**

**Profile, looking south: plant heights step down from the fence toward the orchard**



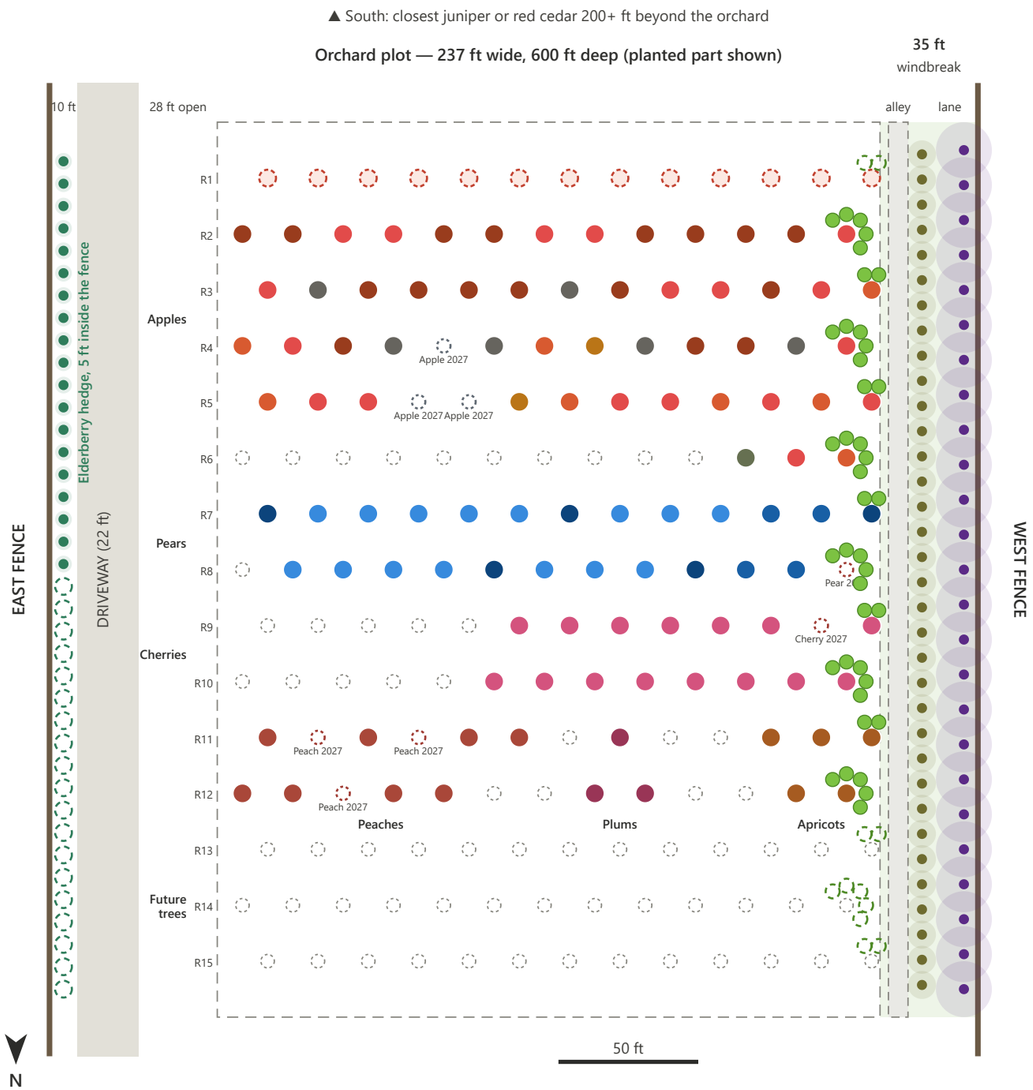




Stretch shown: 14–52 ft north of the R1 line. Green circles: berry guild bushes around the C1 trees (HB honeyberry).

| Layer (west → east) | Plant | Count | From west fence | In-row spacing | First plant | Last plant | Mature size | Notes |
|---|---|---|---|---|---|---|---|---|
| 1. Tall layer on the west fence | **Red mulberry**<br>Morus rubra | 25 | 5 ft | 12.5 ft (12 ft 6 in) | 10 ft south of the R1 line | 10 ft north of the R15 line | 35–50 ft tall; 35–40 ft wide naturally; keep about 20 ft | Main wind barrier. 12.5 ft is within the 6–15 ft in-row range UNL gives for deciduous windbreak trees. It is a tree, usually 15–70 ft tall (USDA Forest Service), but grown in the open it tends to have a short trunk with large, low, spreading limbs (Purdue) and it sprouts from the roots (USDA Forest Service), so left alone it can look like a wide, many-stemmed bush. Seedlings are usually male or female, so expect some fruitless trees (MOBOT). Red and white mulberry cross and are hard to tell apart (Purdue); white (Russian) mulberry, long planted in western windbreaks, is often a smaller, many-branched tree. Prune the east side back to about 10 ft from the trunk so it stays over the walking lane, not the hazelnuts. Ripe mulberries may also draw birds away from the cherries. |
| 2. Shrub layer beside the alley | **Hazelnut**<br>Corylus hybrids | 34 | 20 ft | 9 ft | 8.5 ft south of the R1 line | 8.5 ft north of the R15 line | Up to about 12 ft tall; nearly as wide as tall; keep about 10 ft | 15 ft from the mulberries; USDA's hybrid hazelnut guide keeps taller tree rows at least 15 ft away so they do not shade out the hazels. Hybrids grow as multi-stemmed bushes, not trees, up to about 12 ft tall and nearly as wide, which can take ten years or more (UMHDI). 9 ft apart is within the 6–9 ft UMHDI gives for nut production; layer low stems into the gaps to close the row sooner. Pick by hand as soon as the nuts loosen in their husks (UMHDI); seedlings ripen over a month or more, late August into October (UMHDI harvest bulletin), so plan several passes. After about 12 years, cut a few bushes to the ground each winter to keep them small enough to pick (UMHDI). |
| East hedge beside the driveway | **American elderberry**<br>Sambucus canadensis | 19 planted + 19 to add | 327 ft (5 ft inside the east fence) | 8 ft | 6 ft south of the R1 line | 138 ft north of the R1 line (last planted); open positions to 10 ft north of the R15 line | 6–12 ft tall; 6–8 ft or wider; keep about 6 ft | A separate, tighter hedge in full sun between the driveway and the east fence. WVU suggests at least 8–9 ft apart in the row for fruit production (plants with a 6–8 ft spread). Several Berry-sheet elderberries need another variety, so mix the named cultivars along the row. Fill the open positions with dormant hardwood cuttings, which WVU says root easily. Prune to about 6 ft wide so it stays 2 ft off the driveway, mow suckers, and keep plowed snow off young plants. At 6–12 ft tall it shelters roughly 30–60 ft to the west (about 5 times its height, UNL): the driveway, the headland, and at full height the orchard's east edge, 55 ft away. |

## Site Plan — To Scale, South Up

To scale, south up and west on the right, matching the planting map (the view from the north end looking south). East–west distances are from the west fence; north–south distances are from the R1 row line. Large dots are orchard trees in fruit-type colors; the olive dot at R6-C3 is the planted M-111 rootstock. Filled dashed red rings are positions reserved for grafted rootstock apples, and dashed gray circles are open positions. The unfilled dark-red rings at R8-C1, R9-C2, R11-C10, R11-C12, R12-C11 mark the open positions of the dead trees (1 pear, 1 cherry, and 3 peaches). Slate rings at R4-C9, R5-C9, R5-C10 mark the open spots the Urban apples left. Those trees are planted 20 ft south of R1, off this orchard plot. All 8 spots are labeled for 2027 replacements. Small dots are windbreak plants (purple red mulberry, olive hazelnut) with their managed mature width shaded; the 10 ft walking lane runs between the two rows. Lime dots beside the C1 trees are the berry guilds, and dashed lime rings are guild positions waiting for the R1 and future-row trees. The green dots between the driveway and the east fence are the elderberry hedge, and its dashed rings mark positions for plants to add later. East of the orchard are 28 ft of open ground for turning at the row ends, the driveway, the hedge, and the east fence. The plot is 600 ft deep; only the planted 300 ft is drawn. The block can slide north or south, but keeping it toward the north end adds distance from the junipers. Hover over a dot for its position.
Lot coordinates: x here is the same as on the site master grid, and a spot N ft north of R1 is at grid y = 600 − N (R1-C1 is at x 38, y 600, cell A13; R15 runs along y 320). The whole-site maps, the homestead plot map, and the water plan all use that grid.




### Windbreak, hedge, and berry guild notes

**Row spacing.** The two rows are 15 ft apart: USDA's hybrid hazelnut guide keeps taller tree rows at least 15 ft from the hazels, and UNL's windbreak guide typically spaces rows 12–16 ft apart. The berry bushes moved out of the strip to make this room and now form guilds around the C1 trees.

**Walking lane.** The 10 ft from the mulberry trunks to the west edge of the hazelnuts (5–15 ft from the fence) is for walking and picking. Train each mulberry to one trunk, cut the root sprouts every year, and raise the crown a little each winter so you can walk under it. Remove low limbs while they are small: mulberry cuts on limbs 2 in or thicker may not heal (Ask Extension), and pruning in late fall or winter avoids bleeding (MOBOT). Mulberries are picked by shaking branches over a tarp (Ask Extension), so keep the lane mowed or mulched and clear. Pick the hazelnuts' west side from the lane and their east side from the tractor alley.

**Tractor alley.** The hazelnut row's east edge is 25 ft from the fence and the first orchard trunks are 38 ft from it, leaving 13 ft at ground level. Keep the alley-side limbs of the row-end orchard trees within 6 ft of the trunk to hold a 7 ft clear lane. If the hazelnuts reach their full 12 ft width, the alley narrows to 6 ft (your minimum is 6 ft). C1 trees in the even rows stand another 9 ft east.

**Protection.** UNL reports that a moderately dense windbreak cuts a 20 mph wind to about 5 mph out to 5 times its height. Mature mulberries (35–50 ft) would reach roughly 175–250 ft into the 237 ft orchard; in the first years the hazelnuts mainly shelter the west end. Leafless rows are thinner in early spring, when the stone fruit bloom.

**Length.** Wind wraps around the ends, so UNL recommends extending a windbreak 100–200 ft beyond the area it protects. The mulberry and hazelnut rows cover R1–R15 plus at least 6 ft at each end. The east hedge starts at the south end; 19 more elderberries at the same spacing finish it to the north end. The plot has room to extend all the rows north and south later.

**Junipers.** Junipers and Eastern red cedar host cedar–apple rust, and UMN advises keeping them a few hundred yards from apples; spores can travel more than a mile. Yours are south of the orchard, the closest 200+ ft away, and the apples are at the south end as requested: R1 is about 200+ ft from the closest juniper, and the pears (occasional hosts of a related rust) about 320+ ft. UMN suggests inspecting nearby junipers and red cedars in late winter or early spring and pruning out brown, woody galls before the orange, gelatinous structures form, and notes that the rust is only a minor problem on resistant or partially resistant apples. Do not add junipers or red cedars to this windbreak, and watch the apples for orange leaf spots in late spring.

**Pollination.** Hazelnuts are wind-pollinated and cannot pollinate themselves (UMHDI); the seedling hybrids should pollinate one another. Several named elderberries need another variety (Berry sheet), so mix the cultivars along the hedge.

**Berry guilds.** Each C1 tree in R2–R12 has berry bushes on its alley side: an arc of 5 around the south and west of each even-row tree, which stands 9 ft farther from the alley, and 2 on the south side of each odd-row tree. They sit 6–7 ft from the trunks and about 5 ft apart, all east of the alley (the nearest bush edge is 33 ft from the fence), with a walking gap of at least 4.5 ft between neighboring guilds. South to north: honeyberry ×11 (R2–R4), red currant ×7 (R4–R6), gooseberry ×3 (R6–R7), Viking aronia ×2 (R8), jostaberry ×17 (R8–R12). Left over: 1 jostaberry. The 11 dashed positions around the C1 spots in R1, R13, R14, and R15 wait for those trees. The south side of each tree gets the most sun. Honeyberries tolerate partial shade and need at least two varieties that pollinate each other (UAF Extension), so mix their varieties; currants and gooseberries tolerate partial shade but fruit more in full sun (UMN). Comfrey could fill the east side of each guild, in the tree row, but its deep roots make it difficult to eradicate once established (WSU). Keep one side of each tree open for a ladder.

**Blister rust.** White pine blister rust attacks five-needle (white) pines and uses currants and gooseberries as alternate hosts (WSU). Your white pines died, and ponderosa and Austrian pines are not five-needle pines, so the guild currants and gooseberries put no pines at risk; UMN also lists jostaberry as resistant. Avoid planting white pines near them later.

## Metal Tag to Orchard Position — Planting Checklist

Start at the south end of the orchard with R1 and work north. Within each row, start at C1 on the west/windbreak side and work east toward C13. Match the exact metal **Tag ID** first; the tree name and inventory row are cross-checks. When several trees share one Tag ID, the sequence identifies each otherwise identical inventory instance.
**Spot SiteRef:** `CELL-Eoffset-Soffset` locates the listed orchard spot. Both offsets are feet from the master cell's northwest corner; `B11-E33-S0` is the M-111 at R6-C3. E and S are compass directions, not page directions on this south-up map. Codes stay with vacant spots for 2027 replacements; the Urban apples' new outside-orchard locations remain `PENDING`. See the master-grid guide.

| Planted | Grid position | Spot SiteRef | Metal Tag ID | Code | Tree | Inventory row |
|---|---|---|---|---|---|---|
| ☑ | **R2-C1** | `A12-E47-S30` | **AL-Foxwhelp** | **FOX** | **Foxwhelp Cider Apple**<br>Inventory name: Foxwhelp Cider<br>Late; crimson-striped; bittersharp cider apple | 39 |
| ☑ | **R2-C2** | `B12-E15-S30` | **AL-Brae Dwf** | **BB** | **Braeburn**<br>Inventory name: Braeburn Dwarf<br>Late; red-orange streaks; crisp, tangy-sweet | 27 |
| ☑ | **R2-C3** | `B12-E33-S30` | **AL-Jon G**<br>Tree 1 of 2 with this tag | **JG** | **Jonagold**<br>Late; large red-over-gold; honeyed sweet-tart; triploid | 53 |
| ☑ | **R2-C4** | `C12-E1-S30` | **AL-Fuji** | **FJ** | **Fuji**<br>Very late; pink-red stripes; sweet, crisp, long keeper | 42 |
| ☑ | **R2-C5** | `C12-E19-S30` | **AL-Hon Cris**<br>Tree 1 of 2 with this tag | **HC** | **Honeycrisp**<br>Mid; red-orange flush; very crisp, sweet | 49 |
| ☑ | **R2-C6** | `C12-E37-S30` | **AL-Cort/ AL-Cortland**<br>Tree 1 of 3 with this tag | **CL** | **Cortland**<br>Mid; red-streaked; white flesh slow to brown | 29 |
| ☑ | **R2-C7** | `D12-E5-S30` | **AL-Pink/ AL-Pink Lady**<br>Tree 1 of 2 with this tag | **PL** | **Pink Lady (Cripps Pink)**<br>Inventory name: Pink Lady<br>Very late; pink blush on yellow; crisp, sweet-tart | 59 |
| ☑ | **R2-C8** | `D12-E23-S30` | **AL-Jon G**<br>Tree 2 of 2 with this tag | **JG** | **Jonagold**<br>Late; large red-over-gold; honeyed sweet-tart; triploid | 53 |
| ☑ | **R2-C9** | `D12-E41-S30` | **AL-Gala** | **GL** | **Gala**<br>Early; orange-red streaks; mild, sweet | 45 |
| ☑ | **R2-C10** | `E12-E9-S30` | **AL-Ark Blk** | **AB** | **Arkansas Black Spur (FT015)**<br>Inventory name: Arkansas Black<br>Very late; dark red-black; tart, firm, long keeper | 26 |
| ☑ | **R2-C11** | `E12-E27-S30` | **AL-O Pip**<br>Tree 1 of 2 with this tag | **CO** | **Cox's Orange Pippin**<br>Inventory name: Cox Orange Pippin<br>Mid; orange-red flush; aromatic, sweet-tart; fussy | 30 |
| ☑ | **R2-C12** | `E12-E45-S30` | **AL-Frank C** | **FC** | **Franklin Cider**<br>Late; russeted green-yellow; bittersharp cider apple | 40 |
| ☑ | **R2-C13** | `F12-E13-S30` | **AL-Frequin** | **FRQ** | **Frequin Rouge Cider Apple**<br>Inventory name: Frequin Rouge<br>Mid; small red-striped; bittersweet French cider apple | 41 |
| ☑ | **R3-C1** | `A12-E38-S10` | **AL-Kerr Crab**<br>Tree 1 of 3 with this tag | **KER** | **Kerr Crab Seedling**<br>Inventory name: Crabapple, Kerr<br>Mid; crimson-maroon crab; crisp, tart-sweet; keeps | 32 |
| ☑ | **R3-C2** | `B12-E6-S10` | **AL-Cort/ AL-Cortland**<br>Tree 2 of 3 with this tag | **CL** | **Cortland**<br>Mid; red-streaked; white flesh slow to brown | 29 |
| ☑ | **R3-C3** | `B12-E24-S10` | **AL-G Orch-7** | **KB** | **Kingston Black**<br>Late; dark red, russeted; bittersharp cider apple | 54 |
| ☑ | **R3-C4** | `B12-E42-S10` | **AL-Crisp** | **MC** | **Mutsu Crispin**<br>Inventory name: Crispin<br>Late; big yellow-green; crisp, honeyed; triploid | 35 |
| ☑ | **R3-C5** | `C12-E10-S10` | **AL-Pink/ AL-Pink Lady**<br>Tree 2 of 2 with this tag | **PL** | **Pink Lady (Cripps Pink)**<br>Inventory name: Pink Lady<br>Very late; pink blush on yellow; crisp, sweet-tart | 59 |
| ☑ | **R3-C6** | `C12-E28-S10` | **AL-Az Fuji**<br>Tree 1 of 2 with this tag | **FA** | **Aztec Fuji**<br>Inventory name: Fuji Aztec<br>Very late; deep pink-red; very sweet, long keeper | 43 |

| Planted | Grid position | Spot SiteRef | Metal Tag ID | Code | Tree | Inventory row |
|---|---|---|---|---|---|---|
| ☑ | **R3-C7** | `C12-E46-S10` | **AL-UnKn**<br>Tree 1 of 6 with this tag | **UNK** | **Unidentified apple**<br>Inventory name: Unknown | 62 |
| ☑ | **R3-C8** | `D12-E14-S10` | **AL-Hon Cris**<br>Tree 2 of 2 with this tag | **HC** | **Honeycrisp**<br>Mid; red-orange flush; very crisp, sweet | 49 |
| ☑ | **R3-C9** | `D12-E32-S10` | **AL-Az Fuji**<br>Tree 2 of 2 with this tag | **FA** | **Aztec Fuji**<br>Inventory name: Fuji Aztec<br>Very late; deep pink-red; very sweet, long keeper | 43 |
| ☑ | **R3-C10** | `E12-E0-S10` | **AL-Gld Rsh** | **GO** | **GoldRush**<br>Inventory name: Goldrush<br>Very late; yellow, russet; tart-sweet, keeps 6 months | 46 |
| ☑ | **R3-C11** | `E12-E18-S10` | **AL-Red Del** | **RD** | **Red Delicious**<br>Inventory name: Delicious, Red<br>Late; tall dark red; sweet, mild, crisp; five bumps | 37 |
| ☑ | **R3-C12** | `E12-E36-S10` | **AL-UnKn**<br>Tree 2 of 6 with this tag | **UNK** | **Unidentified apple**<br>Inventory name: Unknown | 62 |
| ☑ | **R3-C13** | `F12-E4-S10` | **AL-G Orch-111** | **NP** | **Newtown Pippen**<br>Inventory name: Newton Yellow Pippin<br>Very late; dull green-yellow; tart, aromatic winter apple | 58 |
| ☑ | **R4-C1** | `A11-E47-S40` | **AL-O Pip**<br>Tree 2 of 2 with this tag | **CO** | **Cox's Orange Pippin**<br>Inventory name: Cox Orange Pippin<br>Mid; orange-red flush; aromatic, sweet-tart; fussy | 30 |
| ☑ | **R4-C2** | `B11-E15-S40` | **AL-UnKn**<br>Tree 3 of 6 with this tag | **UNK** | **Unidentified apple**<br>Inventory name: Unknown | 62 |
| ☑ | **R4-C3** | `B11-E33-S40` | **AL-Hny Crisp** | **HC** | **Honeycrisp (Super Red)**<br>Inventory name: Honeycrisp, Super Red<br>Mid; red over yellow; very crisp, sweet-tart | 50 |
| ☑ | **R4-C4** | `C11-E1-S40` | **AL-Gold D** | **GD** | **Golden Delicious**<br>Inventory name: Delicious, Golden<br>Late; golden-green; sweet, mild honey; keeps well | 36 |
| ☑ | **R4-C5** | `C11-E19-S40` | **AL-UnKn**<br>Tree 4 of 6 with this tag | **UNK** | **Unidentified apple**<br>Inventory name: Unknown | 62 |
| ☑ | **R4-C6** | `C11-E37-S40` | **AL-Zestar**<br>Tree 1 of 2 with this tag | **ZT** | **Zestar**<br>Earliest; rosy red blush; sweet-tart, brown sugar | 66 |
| ☑ | **R4-C7** | `D11-E5-S40` | **AL-Wickson cr** | **WC** | **Wickson Crab**<br>Inventory name: Crabapple, Wickson<br>Late; golf-ball red crab; very sweet, sharp; early bloom | 33 |
| ☑ | **R4-C8** | `D11-E23-S40` | **AL-UnKn**<br>Tree 5 of 6 with this tag | **UNK** | **Unidentified apple**<br>Inventory name: Unknown | 62 |
| **Open** | **R4-C9** | `D11-E41-S40` | **AL-UrbGrn** | **UG** | **Tandgy green Urban Columnar - NOT IN ORCHARD**<br>Inventory name: Urban Green<br>Planned outside the orchard; current SiteRef: D13-E0-S20. Leave this spot open; plant a new apple tree in 2027 (planned: Haralson). | 64 |
| ☑ | **R4-C10** | `E11-E9-S40` | **AL-UnKn**<br>Tree 6 of 6 with this tag | **UNK** | **Unidentified apple**<br>Inventory name: Unknown | 62 |
| ☑ | **R4-C11** | `E11-E27-S40` | **AL-Fuji** | **AF** | **Auvil Early Fuji**<br>Inventory name: Fuji, Early<br>Mid; red, gold-touched; sweet, low-acid Fuji sport | 44 |
| ☑ | **R4-C12** | `E11-E45-S40` | **AL-Harrison**<br>Tree 1 of 2 with this tag | **HR** | **Harrison**<br>Late; small yellow; sweet, rich cider apple | 48 |
| ☑ | **R4-C13** | `F11-E13-S40` | **AL-Dolgo** | **DLG** | **Dolgo Crabapple**<br>Inventory name: Crabapple, Dolgo<br>Early crab; crimson, oblong; tart, white bloom | 31 |

| Planted | Grid position | Spot SiteRef | Metal Tag ID | Code | Tree | Inventory row |
|---|---|---|---|---|---|---|
| ☑ | **R5-C1** | `A11-E38-S20` | **AL-Campfield** | **CAM** | **Campfield Cider Apple**<br>Inventory name: Campfield Cide<br>Very late; red-striped; bittersweet Newark cider apple | 28 |
| ☑ | **R5-C2** | `B11-E6-S20` | **AL-Idared** | **ID** | **Idared**<br>Late; smooth red; mild sweet-tart; keeps 6 months | 52 |
| ☑ | **R5-C3** | `B11-E24-S20` | **AL-Gold Gem** | **HG** | **Hudson's Golden Gem**<br>Inventory name: Hudsons Golden Gem<br>Late; large russet, pear-shaped; sweet, nutty | 51 |
| ☑ | **R5-C4** | `B11-E42-S20` | **AL-Liberty** | **LB** | **Liberty**<br>Mid; bright red Mac-type; scab-resistant; sweet-tart | 55 |
| ☑ | **R5-C5** | `C11-E10-S20` | **AL- S Six** | **SS** | **Sweet Sixteen**<br>Mid; pink-red stripes; yellow flesh, candy-sweet | 60 |
| ☑ | **R5-C6** | `C11-E28-S20` | **AL-Cort/ AL-Cortland**<br>Tree 3 of 3 with this tag | **CL** | **Cortland**<br>Mid; red-streaked; white flesh slow to brown | 29 |
| ☑ | **R5-C7** | `C11-E46-S20` | **AL-Mc In** | **MT** | **Macintosh**<br>Inventory name: McIntosh<br>Mid; red over green; crisp, wine-like, white flesh | 57 |
| ☑ | **R5-C8** | `D11-E14-S20` | **AL-Zestar**<br>Tree 2 of 2 with this tag | **ZT** | **Zestar**<br>Earliest; rosy red blush; sweet-tart, brown sugar | 66 |
| **Open** | **R5-C9** | `D11-E32-S20` | **AL-UrbR** | **UR** | **Tasty Red Urban Columnar - NOT IN ORCHARD**<br>Inventory name: Urban Red<br>Planned outside the orchard; current SiteRef: C13-E46-S20. Leave this spot open; plant a new apple tree in 2027 (planned: Williams' Pride). | 65 |
| **Open** | **R5-C10** | `E11-E0-S20` | **AL-UrbGld** | **UD** | **Blushing Delight Urban Columnar - NOT IN ORCHARD**<br>Inventory name: Urban Gold<br>Planned outside the orchard; current SiteRef: C13-E42-S20. Leave this spot open; plant a new apple tree in 2027 (planned: Priscilla). | 63 |
| ☑ | **R5-C11** | `E11-E18-S20` | **AL-Harrison**<br>Tree 2 of 2 with this tag | **HR** | **Harrison**<br>Late; small yellow; sweet, rich cider apple | 48 |
| ☑ | **R5-C12** | `E11-E36-S20` | **AL-Crimson** | **CC** | **CrimsonCrisp**<br>Inventory name: Crimson Crisp<br>Mid; bright crimson; very crisp, honeyed; scab-immune | 34 |
| ☑ | **R5-C13** | `F11-E4-S20` | **AL-Kerr Crab**<br>Tree 2 of 3 with this tag | **KER** | **Kerr Crab Seedling**<br>Inventory name: Crabapple, Kerr<br>Mid; crimson-maroon crab; crisp, tart-sweet; keeps | 32 |
| ☑ | **R6-C1** | `A11-E47-S0` | **AL-Kerr Crab**<br>Tree 3 of 3 with this tag | **KER** | **Kerr Crab Seedling**<br>Inventory name: Crabapple, Kerr<br>Mid; crimson-maroon crab; crisp, tart-sweet; keeps | 32 |
| ☑ | **R6-C2** | `B11-E15-S0` | **AL-Gran S** | **GS** | **Granny Smith**<br>Very late; bright green; very tart, long keeper | 47 |
| ☑ | **R6-C3** | `B11-E33-S0` | **AL-Rstk M111**<br>Tree 1 of 12 with this tag | **RS** | **M-111 apple rootstock**<br>Inventory name: M 111 Rootstock<br>Already planted; scion cultivar not yet recorded.<br>Large-tree rootstock; well anchored; tolerates dry soil | 56 |
| ☑ | **R7-C1** | `A10-E38-S30` | **PR- Yw Huffcapp** | **YHC** | **Yellow Huffcapp European Perry Pear**<br>Inventory name: Yellow Huffcapp<br>Mid; small elliptical yellow-green; sharp perry | 110 |
| ☑ | **R7-C2** | `B10-E6-S30` | **PR-Shin Asian** | **SA** | **Shinseiki - Asian**<br>Inventory name: Asian, Shinseiki<br>Mid; yellow skin; crisp, sweet, slightly tart | 95 |
| ☑ | **R7-C3** | `B10-E24-S30` | **PR-Olympic** | **OG** | **Olympic Giant - Asian**<br>Inventory name: Olympic Giant<br>Late; huge brown russet; sweet, crisp, long keeper | 106 |

| Planted | Grid position | Spot SiteRef | Metal Tag ID | Code | Tree | Inventory row |
|---|---|---|---|---|---|---|
| ☑ | **R7-C4** | `B10-E42-S30` | **PR-R Bart** | **RB** | **Red Bartlett**<br>Inventory name: Bartlett, Red<br>Mid; red-skinned Bartlett sport; sweet, juicy | 98 |
| ☑ | **R7-C5** | `C10-E10-S30` | **PR-Anjou** | **AU** | **Anjou**<br>Inventory name: Anjou Stan. Supreme<br>Late; green skin, blunt shape; mild, buttery; long keeper | 92 |
| ☑ | **R7-C6** | `C10-E28-S30` | **PR-Comice**<br>Tree 1 of 3 with this tag | **CE** | **Comice**<br>Late; squat, green-yellow, red blush; buttery, aromatic | 100 |
| ☑ | **R7-C7** | `C10-E46-S30` | **PR-Barnet** | **BNT** | **Barnett European Perry Pear**<br>Inventory name: Barnet Euro<br>Late; small, russeted all over; light perry | 96 |
| ☑ | **R7-C8** | `D10-E14-S30` | **PR-Sum Cris**<br>Tree 1 of 2 with this tag | **SMC** | **SummerCrisp**<br>Inventory name: Summercrisp<br>Early; green, red blush; crisp, mild-sweet | 109 |
| ☑ | **R7-C9** | `D10-E32-S30` | **PR-Parker** | **PKR** | **Parker**<br>Mid; yellow-bronze, red blush; hardy, tender-juicy | 107 |
| ☑ | **R7-C10** | `E10-E0-S30` | **PR-Sum Cris**<br>Tree 2 of 2 with this tag | **SMC** | **SummerCrisp**<br>Inventory name: Summercrisp<br>Early; green, red blush; crisp, mild-sweet | 109 |
| ☑ | **R7-C11** | `E10-E18-S30` | **PR-Comice**<br>Tree 2 of 3 with this tag | **CE** | **Comice**<br>Late; squat, green-yellow, red blush; buttery, aromatic | 100 |
| ☑ | **R7-C12** | `E10-E36-S30` | **PR-Kief** | **KF** | **Keiffer**<br>Inventory name: Kieffer<br>Late; yellow, crimson blush; crisp, coarse; canning pear | 104 |
| ☑ | **R7-C13** | `F10-E4-S30` | **PR-Butt Euro** | **BUT** | **Butt European Perry Pear**<br>Inventory name: Butt Euro Perry<br>Late; small lemon-yellow; astringent perry pear | 99 |
| **Open** | **R8-C1** | `A10-E47-S10` | **PR-Pineap** | **PE** | **Pineapple - DEAD**<br>Leave this spot open; plant a new pear tree in 2027 (planned: Harrow Delight). | 108 |
| ☑ | **R8-C2** | `B10-E15-S10` | **PR-Asian H** | **HA** | **Hosui - Asian**<br>Inventory name: Asian Hosui<br>Mid; round golden-russet; sweet-tart, very juicy | 93 |
| ☑ | **R8-C3** | `B10-E33-S10` | **PR-Asian** | **HG** | **Starking Hardy Giant - Asian**<br>Inventory name: Asian, Hardy Giant<br>Late; huge round brown russet; crisp, long keeper | 94 |
| ☑ | **R8-C4** | `C10-E1-S10` | **PR-Hendre**<br>Tree 1 of 2 with this tag | **HHC** | **Hendre Huffcapp European Perry Pear**<br>Inventory name: Hendre Huffeap<br>Late; small elliptical; light low-tannin perry | 103 |
| ☑ | **R8-C5** | `C10-E19-S10` | **PR-Bart**<br>Tree 1 of 2 with this tag | **BT** | **Bartlett**<br>Mid; green turning yellow; sweet, juicy, aromatic | 97 |
| ☑ | **R8-C6** | `C10-E37-S10` | **PR-Moon** | **MW** | **Moonglow**<br>Mid; yellow, pink blush; soft, juicy, mild | 105 |
| ☑ | **R8-C7** | `D10-E5-S10` | **PR-D'an**<br>Tree 1 of 2 with this tag | **AU** | **Anjou**<br>Inventory name: D'Anjou<br>Late; green skin, blunt shape; mild, buttery; long keeper | 101 |
| ☑ | **R8-C8** | `D10-E23-S10` | **PR-Hendre**<br>Tree 2 of 2 with this tag | **HHC** | **Hendre Huffcapp European Perry Pear**<br>Inventory name: Hendre Huffeap<br>Late; small elliptical; light low-tannin perry | 103 |
| ☑ | **R8-C9** | `D10-E41-S10` | **PR-Delic** | **SD** | **Starking Delicious**<br>Inventory name: Delicious<br>Mid; large yellow-green; sweet, juicy, easy to peel | 102 |
| ☑ | **R8-C10** | `E10-E9-S10` | **PR-D'an**<br>Tree 2 of 2 with this tag | **AU** | **Anjou**<br>Inventory name: D'Anjou<br>Late; green skin, blunt shape; mild, buttery; long keeper | 101 |

| Planted | Grid position | Spot SiteRef | Metal Tag ID | Code | Tree | Inventory row |
|---|---|---|---|---|---|---|
| ☑ | **R8-C11** | `E10-E27-S10` | **PR-Bart**<br>Tree 2 of 2 with this tag | **BT** | **Bartlett**<br>Mid; green turning yellow; sweet, juicy, aromatic | 97 |
| ☑ | **R8-C12** | `E10-E45-S10` | **PR-Comice**<br>Tree 3 of 3 with this tag | **CE** | **Comice**<br>Late; squat, green-yellow, red blush; buttery, aromatic | 100 |
| ☑ | **R9-C1** | `A9-E38-S40` | **CR-Bing**<br>Tree 1 of 2 with this tag | **BB** | **Bing**<br>Mid; mahogany to near-black heart; very sweet, firm | 72 |
| **Open** | **R9-C2** | `B9-E6-S40` | **CR-Royalton** | **RS** | **Royalton Sweet - DEAD**<br>Leave this spot open; plant a new cherry tree in 2027 (planned: Black Pearl). | 80 |
| ☑ | **R9-C3** | `B9-E24-S40` | **CR-Bing**<br>Tree 2 of 2 with this tag | **BB** | **Bing**<br>Mid; mahogany to near-black heart; very sweet, firm | 72 |
| ☑ | **R9-C4** | `B9-E42-S40` | **CR-Stella** | **ST** | **Stella**<br>Inventory name: Sweet Stella<br>Mid; self-fertile; large dark heart-shaped, firm | 82 |
| ☑ | **R9-C5** | `C9-E10-S40` | **CR-Ian Nan**<br>Tree 1 of 2 with this tag | **NKI** | **Nanking (bush)**<br>Inventory name: Nanking, Ian<br>Mid; small tart-sweet cherries; very hardy bush | 78 |
| ☑ | **R9-C6** | `C9-E28-S40` | **CR-Sweet**<br>Tree 1 of 3 with this tag | **SWT** | **Unidentified sweet cherry**<br>Inventory name: Sweet | 81 |
| ☑ | **R9-C7** | `C9-E46-S40` | **CR-Hardy Antq** | **HG** | **Hardy Giant Antque Sweet**<br>Inventory name: Hardy Giant Antique<br>Mid; mahogany skin, dark red flesh; heavy clusters | 75 |
| ☑ | **R9-C8** | `D9-E14-S40` | **CR-Ian Nan**<br>Tree 2 of 2 with this tag | **NKI** | **Nanking (bush)**<br>Inventory name: Nanking, Ian<br>Mid; small tart-sweet cherries; very hardy bush | 78 |
| ☑ | **R10-C1** | `A9-E47-S20` | **CR-Black T**<br>Tree 1 of 2 with this tag | **BT** | **Black Tartarian**<br>Early; purple-black, medium; sweet, rich; pollinizer | 73 |
| ☑ | **R10-C2** | `B9-E15-S20` | **CR-Lapins** | **LP** | **Lapins**<br>Inventory name: Lapins Sweet<br>Late; self-fertile; large mahogany, Bing-shaped | 76 |
| ☑ | **R10-C3** | `B9-E33-S20` | **CR-Black T**<br>Tree 2 of 2 with this tag | **BT** | **Black Tartarian**<br>Early; purple-black, medium; sweet, rich; pollinizer | 73 |
| ☑ | **R10-C4** | `C9-E1-S20` | **CR-Sweet**<br>Tree 2 of 3 with this tag | **SWT** | **Unidentified sweet cherry**<br>Inventory name: Sweet | 81 |
| ☑ | **R10-C5** | `C9-E19-S20` | **CR-Mont**<br>Tree 1 of 2 with this tag | **MNT** | **Montmorency**<br>Late; bright red tart pie cherry; self-fertile | 77 |
| ☑ | **R10-C6** | `C9-E37-S20` | **CR-Juls Nan** | **NKJ** | **Nanking (bush)**<br>Inventory name: Nanking, Jules<br>Mid; small tart-sweet cherries; very hardy bush | 79 |
| ☑ | **R10-C7** | `D9-E5-S20` | **CR-Sweet**<br>Tree 3 of 3 with this tag | **SWT** | **Unidentified sweet cherry**<br>Inventory name: Sweet | 81 |
| ☑ | **R10-C8** | `D9-E23-S20` | **CR-Mont**<br>Tree 2 of 2 with this tag | **MNT** | **Montmorency**<br>Late; bright red tart pie cherry; self-fertile | 77 |
| ☑ | **R11-C1** | `A9-E38-S0` | **AT-moorpark**<br>Tree 1 of 2 with this tag | **MP** | **Moorpark**<br>Inventory name: Moore Park<br>Mid; large orange, red-specked; rich flavor; fresh, dried | 70 |
| ☑ | **R11-C2** | `B9-E6-S0` | **AT-Patterson** | **PN** | **Patterson**<br>Early; firm, red-streaked yellow; self-fertile; canning | 71 |
| ☑ | **R11-C3** | `B9-E24-S0` | **AT-Tilton** | **GT** | **Giant Tilton**<br>Mid; large golden-orange; fresh, canning, drying | 68 |
| ☑ | **R11-C6** | `C9-E28-S0` | **PM- It Lov**<br>Tree 1 of 3 with this tag | **IPL** | **Italian Plum**<br>Inventory name: Italian Lovell<br>Mid; dark blue prune plum; sweet, freestone, dries well | 111 |

| Planted | Grid position | Spot SiteRef | Metal Tag ID | Code | Tree | Inventory row |
|---|---|---|---|---|---|---|
| ☑ | **R11-C8** | `D9-E14-S0` | **PC-Contender** | **CR** | **Contender**<br>Mid; red-orange skin; very hardy; non-browning flesh | 85 |
| ☑ | **R11-C9** | `D9-E32-S0` | **PC-Rd Haven** | **RH** | **Redhaven**<br>Early; red-blushed, little fuzz; sweet yellow flesh | 89 |
| **Open** | **R11-C10** | `E9-E0-S0` | **PC-Hale** | **JH** | **JH Hale - DEAD**<br>Inventory name: Hale<br>Leave this spot open; plant a new peach tree in 2027 (planned: Intrepid). | 88 |
| ☑ | **R11-C11** | `E9-E18-S0` | **PC-Frost** | **FT** | **Frost**<br>Mid; leaf curl resistant; yellow-green skin, pink bloom | 87 |
| **Open** | **R11-C12** | `E9-E36-S0` | **PC-Can Harm** | **CH** | **Canadian Harmony - DEAD**<br>Leave this spot open; plant a new peach tree in 2027 (planned: Contender). | 84 |
| ☑ | **R11-C13** | `F9-E4-S0` | **PC-Reliance**<br>Tree 1 of 2 with this tag | **RC** | **Reliance**<br>Early; very hardy, late bloom; soft, sweet, juicy | 90 |
| ☑ | **R12-C1** | `A8-E47-S30` | **AT-Harcot** | **HT** | **Harcot**<br>Mid; orange, red blush; late bloom, canker resistant | 69 |
| ☑ | **R12-C2** | `B8-E15-S30` | **AT-moorpark**<br>Tree 2 of 2 with this tag | **MP** | **Moorpark**<br>Inventory name: Moore Park<br>Mid; large orange, red-specked; rich flavor; fresh, dried | 70 |
| ☑ | **R12-C5** | `C8-E19-S30` | **PM- It Lov**<br>Tree 2 of 3 with this tag | **IPL** | **Italian Plum**<br>Inventory name: Italian Lovell<br>Mid; dark blue prune plum; sweet, freestone, dries well | 111 |
| ☑ | **R12-C6** | `C8-E37-S30` | **PM- It Lov**<br>Tree 3 of 3 with this tag | **IPL** | **Italian Plum**<br>Inventory name: Italian Lovell<br>Mid; dark blue prune plum; sweet, freestone, dries well | 111 |
| ☑ | **R12-C9** | `D8-E41-S30` | **PC-Reliance**<br>Tree 2 of 2 with this tag | **RC** | **Reliance**<br>Early; very hardy, late bloom; soft, sweet, juicy | 90 |
| ☑ | **R12-C10** | `E8-E9-S30` | **PC-Unk**<br>Tree 1 of 2 with this tag | **UNK** | **Unidentified peach**<br>Inventory name: Unknown | 91 |
| **Open** | **R12-C11** | `E8-E27-S30` | **PC-Blaz Star** | **BS** | **BlazingStar - DEAD**<br>Inventory name: Blazing Star<br>Leave this spot open; plant a new peach tree in 2027 (planned: Harrow Diamond). | 83 |
| ☑ | **R12-C12** | `E8-E45-S30` | **PC-Unk**<br>Tree 2 of 2 with this tag | **UNK** | **Unidentified peach**<br>Inventory name: Unknown | 91 |
| ☑ | **R12-C13** | `F8-E13-S30` | **PC-Elberta** | **EA** | **Elberta**<br>Late; large golden-yellow; sweet, firm freestone | 86 |

## Planting Updates — October 4–11, 2026

- R4-C9: Tandgy green Urban Columnar (tag AL-UrbGrn, inventory row 64) not planted in the orchard. This living tree is planted outside the orchard; SiteRef D13-E0-S20. Leave this spot open; the plan is a new Haralson (apple) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock M.7, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan.
- R5-C9: Tasty Red Urban Columnar (tag AL-UrbR, inventory row 65) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E46-S20. Leave this spot open; the plan is a new Williams' Pride (apple) in 2027. It is ordered and has not arrived: Cummins Nursery order 039970, 2026-10-27, ships to Idaho. Rootstock B.118, grade 1, Spring 2027. Described in the planting plan.
- R5-C10: Blushing Delight Urban Columnar (tag AL-UrbGld, inventory row 63) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E42-S20. Leave this spot open; the plan is a new Priscilla (apple) in 2027. Described in the planting plan.
- R8-C1: Pineapple (tag PR-Pineap, inventory row 108) reported dead October 11, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Moonglow, stand-in for Harrow Delight (pear) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock OHxF 97, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan.
- R9-C2: Royalton Sweet (tag CR-Royalton, inventory row 80) reported dead October 17, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Black Pearl (cherry) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order XFEEV5ANY, 2026-10-28, ships to Idaho. Rootstock Mazzard, grade Hand Select, Spring 2027. Described in the planting plan.
- R11-C10: JH Hale (tag PC-Hale, inventory row 88) reported dead October 17, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Intrepid (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Seedling, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan.
- R11-C12: Canadian Harmony (tag PC-Can Harm, inventory row 84) reported dead October 18, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Contender (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Lovell, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan.
- R12-C11: BlazingStar (tag PC-Blaz Star, inventory row 83) reported dead October 18, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Redhaven, stand-in for Harrow Diamond (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Seedling, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan.
- M-111 apple rootstock planted at R6-C3. No scion cultivar is recorded; not counted as a fruit tree or pollen partner until grafting and cultivar identification are recorded.
- Planned from what is at Prairie: 106 living fruit trees in the orchard, plus 1 planted apple rootstock. 3 living Urban apples are planted 20 ft south of R1, outside the orchard. 11 unidentified trees are placed with their fruit type but not counted as pollen donors. The September plan used the full purchase list (122 fruit trees and 32 nut trees).
- Orchard plot set to 237 × 600 ft: the measured 232 ft plus 5 ft from the open ground to the east, so the last trees have 9 ft (half the tree spacing) to the east line. About 28 ft remains between the orchard and the driveway for turning. The September plan used 225 ft plus a 5 ft buffer.
- A 35 ft windbreak strip was added between the orchard and the west fence without shrinking the orchard: rows of red mulberry and hazelnut step down from west to east, followed by a 7 ft clear tractor alley.
- The elderberries moved out of the west windbreak to their own hedge 5 ft inside the east fence, beside the driveway: 8 ft apart for a tighter grouping and some shelter from east winds. The 19 plants fill the south 144 ft; 19 more positions are marked to finish the hedge.
- To leave room to walk and pick, the west strip is now two rows: red mulberry 5 ft and hazelnut 20 ft from the fence, 15 ft apart as USDA's hybrid hazelnut guide advises, with a 10 ft walking lane under the mulberries.
- The berry bushes moved out of the west strip into guilds on the alley side of the C1 trees in R2–R12, following your sketch: 40 bushes (honeyberry ×11, red currant ×7, gooseberry ×3, Viking aronia ×2, jostaberry ×17), with 1 left over. 11 more positions wait for the trees in R1 and the future rows.
- The two white pines died (owner report), so white pine blister rust, which alternates between five-needle pines and currants or gooseberries, is no longer a concern for the currants and gooseberries; ponderosa and Austrian pines are not five-needle pines.
- The main orchard holds fruit trees only. Nut trees, rootstocks, and the two Boise trees are listed separately, and the hazelnuts moved to the windbreak.
- Rows start at the west alley and fill eastward, so open positions are at the east ends, except that in R11–R12 the apricots are right-justified, plums are placed at C6 in R11 and C5–C6 in R12, and peaches are left-justified, so the open spots there are between the blocks, and at R4-C9, R5-C9, R5-C10, R8-C1, R9-C2, R11-C10, R11-C12, R12-C11 after the owner's planting updates.
- Rows by fruit type: apples in 5 rows (13 + 13 + 12 + 11 + 3) plus reserved R1 (13 spots); pears in 2 rows (13 + 11); cherries in 2 rows (7 + 8); apricots at the west end (3 + 2), plums at C6 in R11 and C5–C6 in R12 (1 + 2), and peaches at the east end (4 + 4) share R11–R12; R13–R15 are open for trees you buy or graft later. The cherries use the two rows you chose, and the shared rows R11–R12 follow your layout; the orchard keeps all 15 rows.
- Added R1, a full apple row at the top of the map with 13 spots reserved for grafted rootstock apples. 14 rootstocks remain available (2 EMLA 7, 11 M-111, 1 unknown), after the M-111 planted at R6-C3.
- The fruit types run south to north as apples, pears, cherries, then apricots, plums, and peaches sharing R11–R12, with the apples at the top of the map; the September plan had apples, pears, cherries, apricots, peaches, and plums, each in its own rows. The apples are about 200+ ft from the closest juniper, closer than UMN advises; the windbreak notes list ways to limit cedar–apple rust.
- The J.H. Hale peach died (owner report, Oct 17); the identified peaches that remain are all self-fertile.
- The map is drawn with south at the top and the west fence on the right, the view from the north end looking south; the site plan and windbreak cross-section match. The earlier grid put south at the top but kept west on the left, which mirrored the real layout.

## Prairie Inventory vs. Purchase List

In the orchard: 106 living fruit trees (11 unidentified) plus 1 planted M-111 rootstock. Also at Prairie: 3 living Urban apples planted outside the orchard, 20 ft south of R1, 14 rootstocks still available for grafting and planting, 20 nut trees, 59 west windbreak plants, 19 elderberries for the east hedge, and 41 berry bushes for the C1 guilds. The purchase comparison uses the October 4, 2026 inventory snapshot; later owner planting updates are recorded separately, not treated as missing or unidentified trees. Compared with the Trees.xlsx Fruit sheet, 21 purchased fruit trees were not found at Prairie, 4 more than purchased were found for a few varieties, and 2 trees remain in Boise. Some missing trees may be among the unidentified ones. The table lists only the differences; the line-by-line match is below it.

### Purchased vs. found in the October 4 inventory — differences only

| Purchase record(s), Trees.xlsx Fruit sheet | Purchased | At Prairie | Elsewhere | Difference | Notes |
|---|---|---|---|---|---|
| **Aztec Fuji**<br>Apple; Fruit rows 4, 5 | 3 | 2 | | -1 | 2 of 3 purchased found (row 4: 1 semi-dwarf; row 5: 2 dwarf); the inventory does not say which. May be among the 6 unidentified apples at Prairie. |
| **Braeburn**<br>Apple; Fruit rows 7, 8 | 2 | 1 | | -1 | 1 of 2 purchased Braeburns found. The tag says Dwarf, so it is matched to the Lowes tree with no recorded rootstock (row 8) rather than the M-111 tree (row 7). May be among the 6 unidentified apples at Prairie. |
| **Cortland**<br>Apple; Fruit row 10 | 5 | 3 | | -2 | 3 of 5 purchased found. May be among the 6 unidentified apples at Prairie. |
| **Dolgo Crabapple**<br>Apple; Fruit row 13 | 2 | 1 | | -1 | 1 of 2 purchased found. May be among the 6 unidentified apples at Prairie. |

| Purchase record(s), Trees.xlsx Fruit sheet | Purchased | At Prairie | Elsewhere | Difference | Notes |
|---|---|---|---|---|---|
| **Empire**<br>Apple; Fruit row 14 | 1 | 0 | | -1 | Not in the Prairie inventory. May be among the 6 unidentified apples at Prairie. |
| **Gala**<br>Apple; Fruit rows 19, 20 | 2 | 1 | | -1 | 1 of 2 purchased found. May be among the 6 unidentified apples at Prairie. |
| **Golden Russet**<br>Apple; Fruit row 22 | 1 | 0 | | -1 | Not in the Prairie inventory. May be among the 6 unidentified apples at Prairie. |
| **Honeycrisp**<br>Apple; Fruit row 28 | 3 | 2 | | -1 | 2 of 3 purchased found. May be among the 6 unidentified apples at Prairie. |
| **Kerr Crab Seedling**<br>Apple; Fruit row 33 | 2 | 3 | | +1 | 3 found, but 2 were purchased. All are seed-grown, so each may bloom differently. |
| **Sweet Sixteen**<br>Apple; Fruit row 41 | 2 | 1 | | -1 | 1 of 2 purchased found. May be among the 6 unidentified apples at Prairie. |
| **Zestar**<br>Apple; Fruit row 45 | 3 | 2 | | -1 | 2 of 3 purchased found. May be among the 6 unidentified apples at Prairie. |
| **Chinese (Mormon)**<br>Apricot; Fruit row 46 | 1 | 0 | 1 | 0 | Still in Boise (girdled, mulched); not planned for Prairie. |
| **Moorpark**<br>Apricot; Fruit row 49 | 1 | 2 | | +1 | 2 found, but 1 Moorpark was purchased. The extra Moorpark tag is still unexplained. The former AT-Unk tree (row 71) was identified as Patterson on 2026-10-25, so it is not one of these. |
| **Black Tartarian**<br>Cherry; Fruit rows 55, 56 | 4 | 2 | | -2 | 2 of 4 purchased found. May be among the 3 unidentified sweet cherries at Prairie. |
| **Craig's Crimson**<br>Cherry; Fruit row 57 | 1 | 0 | 1 | 0 | Still in Boise (girdled, mulched); not planned for Prairie. |
| **Regina**<br>Cherry; Fruit row 62 | 1 | 0 | | -1 | Not in the Prairie inventory. May be among the 3 unidentified sweet cherries at Prairie. |
| **Sam Sweet**<br>Cherry; Fruit row 64 | 1 | 0 | | -1 | Not in the Prairie inventory. May be among the 3 unidentified sweet cherries at Prairie. |
| **Hybrid Hazelnut / Jefferson Filbert Hazelnut / Theta Filbert Hazelnut / Hybrid Hazelnut (controlled-cross)**<br>Nut; Fruit rows 70, 71, 76, 118 | 77 | 34 | | -43 | Listed as Jefferson Filbert, but the purchase list has 75 Forestag hybrid seedlings, 1 Jefferson, and 1 Theta. Most are probably hybrids; if the Jefferson and Theta can be identified, plant them side by side (Theta pollinizes Jefferson). |
| **Pecan (Native Hardy)**<br>Nut; Fruit row 74 | 10 | 4 | | -6 | 4 of 10 purchased found. |
| **Shellbark Hickory**<br>Nut; Fruit row 75 | 10 | 0 | | -10 | Not in the Prairie inventory. |
| **Contender**<br>Peach; Fruit row 79 | 2 | 1 | | -1 | 1 of 2 purchased found. May be among the 2 unidentified peaches at Prairie. |
| **Elberta**<br>Peach; Fruit row 80 | 2 | 1 | | -1 | 1 of 2 purchased found. May be among the 2 unidentified peaches at Prairie. |
| **Reliance**<br>Peach; Fruit rows 85, 86, 87 | 3 | 2 | | -1 | 2 of 3 purchased found. May be among the 2 unidentified peaches at Prairie. |
| **Veteran**<br>Peach; Fruit row 88 | 1 | 0 | | -1 | Not in the Prairie inventory. May be among the 2 unidentified peaches at Prairie. |
| **Anjou**<br>Pear; Fruit row 89 | 2 | 3 | | +1 | Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased. |
| **Hendre Huffcapp European Perry Pear**<br>Pear; Fruit row 95 | 1 | 2 | | +1 | 2 found, but 1 was purchased. Thorn (row 107) was not found; one of these may be Thorn. |
| **Thorn European Perry Pear**<br>Pear; Fruit row 107 | 1 | 0 | | -1 | Not in the Prairie inventory. |
| **Italian Plum / Italian Seedling**<br>Plum; Fruit rows 109, 110 | 4 | 3 | | -1 | 3 of 4 found (2 named Italian plums, row 109; 2 Italian seedlings, row 110). Lovell is a peach rootstock, which suggests grafted trees; confirm whether any are seedlings. |
| **Sambucus canadensis**<br>Elderbery; Fruit row 115 | 25 | 19 | | -6 | Planted as a hedge along the east fence by the driveway. The purchase list also has 9 named elderberries on the Berry sheet (American, Bob Gordon ×2, Marge ×2, Perl, Ranch, Wyldewood, York). The 19 at Prairie are not identified by cultivar. |

| Purchase record(s), Trees.xlsx Fruit sheet | Purchased | At Prairie | Elsewhere | Difference | Notes |
|---|---|---|---|---|---|
| **Amelanchier canadensis**<br>Serviceberry; Fruit row 117 | 25 | 0 | | -25 | Not in the Prairie inventory. |

### Show every inventory line used in this plan and where it goes

| Inventory row | Prairie inventory | Qty | Matched purchase record(s) | Plan / positions | Notes |
|---|---|---|---|---|---|
| 3 | **Currant — Red Berry**<br>Tag Currant | 7 | Berry rows 4–7: 5 currants (1 red, 3 black, 1 white) | C1 berry guilds, R4–R6 | 7 currants listed as red at Prairie vs. 5 purchased (1 red, 3 black, 1 white). Self-fertile per the Berry sheet. |
| 4 | **Elderberry** | 19 | Row 115: Sambucus canadensis (25) | East hedge, 5 ft inside the east fence | Planted as a hedge along the east fence by the driveway. The purchase list also has 9 named elderberries on the Berry sheet (American, Bob Gordon ×2, Marge ×2, Perl, Ranch, Wyldewood, York). The 19 at Prairie are not identified by cultivar. |
| 5 | **Gooseberry**<br>Tag GooseBerry | 3 | Berry rows 16–20: 5 gooseberries (Captivator, Hinnonmaki Red ×2, Invicta, Pixwell) | C1 berry guilds, R6–R7 | 3 at Prairie, 2 fewer than purchased. Self-fertile per the Berry sheet. |
| 7 | **Honeyberry**<br>Tag Hny Bry | 11 | Berry rows 26–31: 12 honeyberries (Blue Diamond ×3, Blue Palm ×3, Indigo Treat ×2, Cinderella ×2, Maxine, Sunrise) | C1 berry guilds, R2–R4 | 11 at Prairie, 1 fewer than purchased. The Berry sheet says they need another honeyberry, so plant them as one group and mix the varieties. |
| 8 | **Josta — Giant**<br>Tag JstB-G | 4 | Berry rows 32–34: 7 jostaberries (6 black, 1 red) | C1 berry guilds, R8–R12 (1 left over) | 18 jostaberries at Prairie on lines 8–10, 11 more than purchased. Self-fertile per the Berry sheet. |
| 9 | **Josta — Red**<br>Tag JstB-R | 10 | Berry rows 32–34: 7 jostaberries (6 black, 1 red) | C1 berry guilds, R8–R12 (1 left over) | 18 jostaberries at Prairie on lines 8–10, 11 more than purchased. Self-fertile per the Berry sheet. |
| 10 | **Josta — Regular**<br>Tag JstB | 4 | Berry rows 32–34: 7 jostaberries (6 black, 1 red) | C1 berry guilds, R8–R12 (1 left over) | 18 jostaberries at Prairie on lines 8–10, 11 more than purchased. Self-fertile per the Berry sheet. |
| 11 | **Mullberry — Red** | 25 | Row 116: Morus rubra (25) | West windbreak, red mulberry row | |
| 13 | **Viking**<br>Tag Viking | 2 | Berry rows 2–3: 2 Viking aronia (chokeberry) | C1 berry guilds, R8 | Matches the purchase list. Self-fertile per the Berry sheet. |
| 15 | **Hazlenut — Jefferson Filbert** | 34 | Row 70: Hybrid Hazelnut (25); Row 71: Jefferson Filbert Hazelnut (1); Row 76: Theta Filbert Hazelnut (1); Row 118: Hybrid Hazelnut (controlled-cross) (50) | West windbreak, hazelnut row | Listed as Jefferson Filbert, but the purchase list has 75 Forestag hybrid seedlings, 1 Jefferson, and 1 Theta. Most are probably hybrids; if the Jefferson and Theta can be identified, plant them side by side (Theta pollinizes Jefferson). |
| 26 | **Apple — Arkansas Black**<br>Tag AL-Ark Blk | 1 | Row 2: Arkansas Black Spur (FT015) (1) | R2-C10 | |
| 27 | **Apple — Braeburn Dwarf**<br>Tag AL-Brae Dwf | 1 | Row 8: Braeburn (1); Row 7: Braeburn (1) | R2-C2 | 1 of 2 purchased Braeburns found. The tag says Dwarf, so it is matched to the Lowes tree with no recorded rootstock (row 8) rather than the M-111 tree (row 7). |
| 28 | **Apple — Campfield Cide**<br>Tag AL-Campfield | 1 | Row 9: Campfield Cider Apple (1) | R5-C1 | |
| 29 | **Apple — Cortland**<br>Tag AL-Cort/ AL-Cortland | 3 | Row 10: Cortland (5) | R2-C6, R3-C2, R5-C6 | 3 of 5 purchased found. |
| 30 | **Apple — Cox Orange Pippin**<br>Tag AL-O Pip | 2 | Row 11: Cox's Orange Pippin (2) | R2-C11, R4-C1 | |
| 31 | **Apple — Crabapple, Dolgo**<br>Tag AL-Dolgo | 1 | Row 13: Dolgo Crabapple (2) | R4-C13 | 1 of 2 purchased found. |
| 32 | **Apple — Crabapple, Kerr**<br>Tag AL-Kerr Crab | 3 | Row 33: Kerr Crab Seedling (2) | R3-C1, R5-C13, R6-C1 | 3 found, but 2 were purchased. All are seed-grown, so each may bloom differently. |
| 33 | **Apple — Crabapple, Wickson**<br>Tag AL-Wickson cr | 1 | Row 44: Wickson Crab (1) | R4-C7 | |
| 34 | **Apple — Crimson Crisp**<br>Tag AL-Crimson | 1 | Row 12: CrimsonCrisp (1) | R5-C12 | |
| 35 | **Apple — Crispin**<br>Tag AL-Crisp | 1 | Row 37: Mutsu Crispin (1) | R3-C4 | |
| 36 | **Apple — Delicious, Golden**<br>Tag AL-Gold D | 1 | Row 21: Golden Delicious (1) | R4-C4 | |
| 37 | **Apple — Delicious, Red**<br>Tag AL-Red Del | 1 | Row 40: Red Delicious (1) | R3-C11 | |
| 38 | **Apple — EMLA7 Apple Rootstock**<br>Tag AL-Rtsk-EMLA7 | 2 | No purchase record matched | 2 remain for grafting and planting; Graft first; reserved row R1 holds 13 of them. | Ungrafted EMLA 7 rootstock, for grafting. |

| Inventory row | Prairie inventory | Qty | Matched purchase record(s) | Plan / positions | Notes |
|---|---|---|---|---|---|
| 39 | **Apple — Foxwhelp Cider**<br>Tag AL-Foxwhelp | 1 | Row 15: Foxwhelp Cider Apple (1) | R2-C1 | |
| 40 | **Apple — Franklin Cider**<br>Tag AL-Frank C | 1 | Row 16: Franklin Cider (1) | R2-C12 | |
| 41 | **Apple — Frequin Rouge**<br>Tag AL-Frequin | 1 | Row 17: Frequin Rouge Cider Apple (1) | R2-C13 | |
| 42 | **Apple — Fuji**<br>Tag AL-Fuji | 1 | Row 18: Fuji (1) | R2-C4 | Same tag text (AL-Fuji) as the Early Fuji; check the variety label when planting. |
| 43 | **Apple — Fuji Aztec**<br>Tag AL-Az Fuji | 2 | Row 4: Aztec Fuji (1); Row 5: Aztec Fuji (2) | R3-C6, R3-C9 | 2 of 3 purchased found (row 4: 1 semi-dwarf; row 5: 2 dwarf); the inventory does not say which. |
| 44 | **Apple — Fuji, Early**<br>Tag AL-Fuji | 1 | Row 3: Auvil Early Fuji (1) | R4-C11 | Same tag text (AL-Fuji) as the Fuji; check the variety label when planting. |
| 45 | **Apple — Gala**<br>Tag AL-Gala | 1 | Row 19: Gala (1); Row 20: Gala (1) | R2-C9 | 1 of 2 purchased found. |
| 46 | **Apple — Goldrush**<br>Tag AL-Gld Rsh | 1 | Row 24: GoldRush (1) | R3-C10 | |
| 47 | **Apple — Granny Smith**<br>Tag AL-Gran S | 1 | Row 25: Granny Smith (1) | R6-C2 | |
| 48 | **Apple — Harrison**<br>Tag AL-Harrison | 2 | Row 26: Harrison (1); Row 27: Harrison Cider Apple (1) | R4-C12, R5-C11 | The workbook lists one Harrison (row 26, M-111) and one Harrison Cider (row 27, semi-dwarf), the same cultivar. The tags do not say which is which, so both use the row 26 cultivar data. |
| 49 | **Apple — Honeycrisp**<br>Tag AL-Hon Cris | 2 | Row 28: Honeycrisp (3) | R2-C5, R3-C8 | 2 of 3 purchased found. |
| 50 | **Apple — Honeycrisp, Super Red**<br>Tag AL-Hny Crisp | 1 | Row 29: Honeycrisp (Super Red) (1) | R4-C3 | |
| 51 | **Apple — Hudsons Golden Gem**<br>Tag AL-Gold Gem | 1 | Row 30: Hudson's Golden Gem (1) | R5-C3 | |
| 52 | **Apple — Idared**<br>Tag AL-Idared | 1 | Row 31: Idared (1) | R5-C2 | |
| 53 | **Apple — Jonagold**<br>Tag AL-Jon G | 2 | Row 32: Jonagold (2) | R2-C3, R2-C8 | |
| 54 | **Apple — Kingston Black**<br>Tag AL-G Orch-7 | 1 | Row 34: Kingston Black (1) | R3-C3 | Tag AL-G Orch-7 (Grandpa's Orchard, M-7) matches the workbook. |
| 55 | **Apple — Liberty**<br>Tag AL-Liberty | 1 | Row 35: Liberty (1) | R5-C4 | |
| 56 | **Apple — M 111 Rootstock**<br>Tag AL-Rstk M111 | 12 | No purchase record matched | 11 remain for grafting and planting; Graft first; reserved row R1 holds 13 of them.<br>Already planted: R6-C3 | Ungrafted M-111 rootstock, for grafting. |
| 57 | **Apple — McIntosh**<br>Tag AL-Mc In | 1 | Row 36: Macintosh (1) | R5-C7 | |
| 58 | **Apple — Newton Yellow Pippin**<br>Tag AL-G Orch-111 | 1 | Row 38: Newtown Pippen (1) | R3-C13 | Tag AL-G Orch-111 (Grandpa's Orchard, M-111) matches the workbook. |
| 59 | **Apple — Pink Lady**<br>Tag AL-Pink/ AL-Pink Lady | 2 | Row 39: Pink Lady (Cripps Pink) (2) | R2-C7, R3-C5 | |
| 60 | **Apple — Sweet Sixteen**<br>Tag AL- S Six | 1 | Row 41: Sweet Sixteen (2) | R5-C5 | 1 of 2 purchased found. |
| 61 | **Apple — Ukn Rootstock**<br>Tag AL-Rtsk Unk | 1 | No purchase record matched | 1 remain for grafting and planting; Graft first; reserved row R1 holds 13 of them. | Rootstock of unknown type, for grafting. |
| 62 | **Apple — Unknown**<br>Tag AL-UnKn | 6 | No purchase record matched | R3-C7, R3-C12, R4-C2, R4-C5, R4-C8, R4-C10 | Tag AL-UnKn. Owner (Oct 16, 2026): every purchased apple arrived, so a purchase with no tagged tree is presumed to have died in Boise, unrecorded. These 6 are among the 11 purchased apples with no tagged tree: Braeburn 1, Cortland 2, Dolgo 1 (long shot), Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Identify them at bloom or harvest using the possibility list. |

| Inventory row | Prairie inventory | Qty | Matched purchase record(s) | Plan / positions | Notes |
|---|---|---|---|---|---|
| 63 | **Apple — Urban Gold**<br>Tag AL-UrbGld | 1 | Row 6: Blushing Delight Urban Columnar (1) | R5-C10 - NOT PLANTED HERE; open for a replacement in 2027 | Urban Gold is matched to Blushing Delight, the yellow Urban apple that was purchased (row 6). The workbook's Golden Treat Urban (row 23) has QTY 0; confirm the label. R5-C10: Blushing Delight Urban Columnar (tag AL-UrbGld, inventory row 63) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E42-S20. Leave this spot open; the plan is a new Priscilla (apple) in 2027. Described in the planting plan. |
| 64 | **Apple — Urban Green**<br>Tag AL-UrbGrn | 1 | Row 42: Tandgy green Urban Columnar (1) | R4-C9 - NOT PLANTED HERE; open for a replacement in 2027 | R4-C9: Tandgy green Urban Columnar (tag AL-UrbGrn, inventory row 64) not planted in the orchard. This living tree is planted outside the orchard; SiteRef D13-E0-S20. Leave this spot open; the plan is a new Haralson (apple) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock M.7, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan. |
| 65 | **Apple — Urban Red**<br>Tag AL-UrbR | 1 | Row 43: Tasty Red Urban Columnar (1) | R5-C9 - NOT PLANTED HERE; open for a replacement in 2027 | R5-C9: Tasty Red Urban Columnar (tag AL-UrbR, inventory row 65) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E46-S20. Leave this spot open; the plan is a new Williams' Pride (apple) in 2027. It is ordered and has not arrived: Cummins Nursery order 039970, 2026-10-27, ships to Idaho. Rootstock B.118, grade 1, Spring 2027. Described in the planting plan. |
| 66 | **Apple — Zestar**<br>Tag AL-Zestar | 2 | Row 45: Zestar (3) | R4-C6, R5-C8 | 2 of 3 purchased found. |
| 67 | **Apricot — Chinese (Mormon)**<br>Tag AT-Chinese | 1 | Row 46: Chinese (Mormon) (1) | Not at Prairie (Boise) | Still in Boise (girdled, mulched); not planned for Prairie. |
| 68 | **Apricot — Giant Tilton**<br>Tag AT-Tilton | 1 | Row 47: Giant Tilton (1) | R11-C3 | |
| 69 | **Apricot — Harcot**<br>Tag AT-Harcot | 1 | Row 48: Harcot (1) | R12-C1 | |
| 70 | **Apricot — Moore Park**<br>Tag AT-moorpark | 2 | Row 49: Moorpark (1) | R11-C1, R12-C2 | 2 found, but 1 Moorpark was purchased. The extra Moorpark tag is still unexplained. The former AT-Unk tree (row 71) was identified as Patterson on 2026-10-25, so it is not one of these. |
| 71 | **Apricot — Patterson**<br>Tag AT-Patterson | 1 | Row 50: Patterson (1) | R11-C2 | Owner identified the AT-Unk apricot as Patterson on 2026-10-25. Planted at R11-C2. Metal tag is now AT-Patterson. |
| 72 | **Cherry — Bing**<br>Tag CR-Bing | 2 | Row 53: Bing (1); Row 54: Bing (1) | R9-C1, R9-C3 | |
| 73 | **Cherry — Black Tartarian**<br>Tag CR-Black T | 2 | Row 55: Black Tartarian (3); Row 56: Black Tartarian (1) | R10-C1, R10-C3 | 2 of 4 purchased found. |
| 74 | **Cherry — Craigs Crimson**<br>Tag CR-Crimson | 1 | Row 57: Craig's Crimson (1) | Not at Prairie (Boise) | Still in Boise (girdled, mulched); not planned for Prairie. |
| 75 | **Cherry — Hardy Giant Antique**<br>Tag CR-Hardy Antq | 1 | Row 58: Hardy Giant Antque Sweet (1) | R9-C7 | |
| 76 | **Cherry — Lapins Sweet**<br>Tag CR-Lapins | 1 | Row 59: Lapins (1) | R10-C2 | |
| 77 | **Cherry — Montmorency**<br>Tag CR-Mont | 2 | Row 60: Montmorency (2) | R10-C5, R10-C8 | |
| 78 | **Cherry — Nanking, Ian**<br>Tag CR-Ian Nan | 2 | Row 61: Nanking (bush) (3) | R9-C5, R9-C8 | Nanking 'Ian' (2) and 'Jules' (1) together match the 3 purchased Nanking bushes. |
| 79 | **Cherry — Nanking, Jules**<br>Tag CR-Juls Nan | 1 | Row 61: Nanking (bush) (3) | R10-C6 | Nanking 'Ian' (2) and 'Jules' (1) together match the 3 purchased Nanking bushes. |
| 80 | **Cherry — Royalton Sweet**<br>Tag CR-Royalton | 1 | Row 63: Royalton Sweet (1) | R9-C2 - DEAD (in its pot, before planting); open for a replacement in 2027 | R9-C2: Royalton Sweet (tag CR-Royalton, inventory row 80) reported dead October 17, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Black Pearl (cherry) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order XFEEV5ANY, 2026-10-28, ships to Idaho. Rootstock Mazzard, grade Hand Select, Spring 2027. Described in the planting plan. |
| 81 | **Cherry — Sweet**<br>Tag CR-Sweet | 3 | No purchase record matched | R9-C6, R10-C4, R10-C7 | Tag CR-Sweet. Purchased sweet cherries not otherwise found at Prairie: Black Tartarian 2, Regina 1, Sam 1. |
| 82 | **Cherry — Sweet Stella**<br>Tag CR-Stella | 1 | Row 65: Stella (1) | R9-C4 | |

| Inventory row | Prairie inventory | Qty | Matched purchase record(s) | Plan / positions | Notes |
|---|---|---|---|---|---|
| 83 | **Peach — Blazing Star**<br>Tag PC-Blaz Star | 1 | Row 77: BlazingStar (1) | R12-C11 - DEAD (in its pot, before planting); open for a replacement in 2027 | R12-C11: BlazingStar (tag PC-Blaz Star, inventory row 83) reported dead October 18, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Redhaven, stand-in for Harrow Diamond (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Seedling, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan. |
| 84 | **Peach — Canadian Harmony**<br>Tag PC-Can Harm | 1 | Row 78: Canadian Harmony (1) | R11-C12 - DEAD (in its pot, before planting); open for a replacement in 2027 | R11-C12: Canadian Harmony (tag PC-Can Harm, inventory row 84) reported dead October 18, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Contender (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Lovell, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan. |
| 85 | **Peach — Contender**<br>Tag PC-Contender | 1 | Row 79: Contender (2) | R11-C8 | 1 of 2 purchased found. |
| 86 | **Peach — Elberta**<br>Tag PC-Elberta | 1 | Row 80: Elberta (2) | R12-C13 | 1 of 2 purchased found. |
| 87 | **Peach — Frost**<br>Tag PC-Frost | 1 | Row 81: Frost (1) | R11-C11 | |
| 88 | **Peach — Hale**<br>Tag PC-Hale | 1 | Row 83: JH Hale (1) | R11-C10 - DEAD (in its pot, before planting); open for a replacement in 2027 | R11-C10: JH Hale (tag PC-Hale, inventory row 88) reported dead October 17, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Intrepid (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Seedling, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan. |
| 89 | **Peach — Redhaven**<br>Tag PC-Rd Haven | 1 | Row 84: Redhaven (1) | R11-C9 | |
| 90 | **Peach — Reliance**<br>Tag PC-Reliance | 2 | Row 85: Reliance (1); Row 86: Reliance (1); Row 87: Reliance (1) | R11-C13, R12-C9 | 2 of 3 purchased found. |
| 91 | **Peach — Unknown**<br>Tag PC-Unk | 2 | No purchase record matched | R12-C10, R12-C12 | Tag PC-Unk. Purchased peaches not otherwise found at Prairie: Contender 1, Elberta 1, Reliance 1, Veteran 1. |
| 92 | **Pear — Anjou Stan. Supreme**<br>Tag PR-Anjou | 1 | Row 89: Anjou (2) | R7-C5 | Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased. |
| 93 | **Pear — Asian Hosui**<br>Tag PR-Asian H | 1 | Row 96: Hosui - Asian (1) | R8-C2 | |
| 94 | **Pear — Asian, Hardy Giant**<br>Tag PR-Asian | 1 | Row 105: Starking Hardy Giant - Asian (1) | R8-C3 | |
| 95 | **Pear — Asian, Shinseiki**<br>Tag PR-Shin Asian | 1 | Row 103: Shinseiki - Asian (1) | R7-C2 | |
| 96 | **Pear — Barnet Euro**<br>Tag PR-Barnet | 1 | Row 90: Barnett European Perry Pear (1) | R7-C7 | |
| 97 | **Pear — Bartlett**<br>Tag PR-Bart | 2 | Row 91: Bartlett (1); Row 92: Bartlett (1) | R8-C5, R8-C11 | |
| 98 | **Pear — Bartlett, Red**<br>Tag PR-R Bart | 1 | Row 102: Red Bartlett (1) | R7-C4 | |
| 99 | **Pear — Butt Euro Perry**<br>Tag PR-Butt Euro | 1 | Row 93: Butt European Perry Pear (1) | R7-C13 | |
| 100 | **Pear — Comice**<br>Tag PR-Comice | 3 | Row 94: Comice (3) | R7-C6, R7-C11, R8-C12 | |
| 101 | **Pear — D'Anjou**<br>Tag PR-D'an | 2 | Row 89: Anjou (2) | R8-C7, R8-C10 | Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased. |
| 102 | **Pear — Delicious**<br>Tag PR-Delic | 1 | Row 104: Starking Delicious (1) | R8-C9 | |
| 103 | **Pear — Hendre Huffeap**<br>Tag PR-Hendre | 2 | Row 95: Hendre Huffcapp European Perry Pear (1) | R8-C4, R8-C8 | 2 found, but 1 was purchased. Thorn (row 107) was not found; one of these may be Thorn. |
| 104 | **Pear — Kieffer**<br>Tag PR-Kief | 1 | Row 97: Keiffer (1) | R7-C12 | |

| Inventory row | Prairie inventory | Qty | Matched purchase record(s) | Plan / positions | Notes |
|---|---|---|---|---|---|
| 105 | **Pear — Moonglow**<br>Tag PR-Moon | 1 | Row 98: Moonglow (1) | R8-C6 | |
| 106 | **Pear — Olympic Giant**<br>Tag PR-Olympic | 1 | Row 99: Olympic Giant - Asian (1) | R7-C3 | |
| 107 | **Pear — Parker**<br>Tag PR-Parker | 1 | Row 100: Parker (1) | R7-C9 | |
| 108 | **Pear — Pineapple**<br>Tag PR-Pineap | 1 | Row 101: Pineapple (1) | R8-C1 - DEAD (in its pot, before planting); open for a replacement in 2027 | R8-C1: Pineapple (tag PR-Pineap, inventory row 108) reported dead October 11, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Moonglow, stand-in for Harrow Delight (pear) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock OHxF 97, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan. |
| 109 | **Pear — Summercrisp**<br>Tag PR-Sum Cris | 2 | Row 106: SummerCrisp (2) | R7-C8, R7-C10 | |
| 110 | **Pear — Yellow Huffcapp**<br>Tag PR- Yw Huffcapp | 1 | Row 108: Yellow Huffcapp European Perry Pear (1) | R7-C1 | |
| 111 | **Plum — Italian Lovell**<br>Tag PM- It Lov | 3 | Row 109: Italian Plum (2); Row 110: Italian Seedling (2) | R11-C6, R12-C5, R12-C6 | 3 of 4 found (2 named Italian plums, row 109; 2 Italian seedlings, row 110). Lovell is a peach rootstock, which suggests grafted trees; confirm whether any are seedlings. |
| 112 | **Hican**<br>Tag hican | 10 | Row 69: Hican (10) | Not in the main orchard (nut tree) | |
| 113 | **Pecan** | 4 | Row 74: Pecan (Native Hardy) (10) | Not in the main orchard (nut tree) | 4 of 10 purchased found. |
| 114 | **Walnut — English**<br>Tag WL-Enligsh | 4 | No purchase record matched | Not in the main orchard (nut tree) | Not on the Fruit sheet. The inventory notes 3 are already planted. |
| 115 | **Walnut — Heartnut Japanese**<br>Tag WL-Heart J | 2 | Row 68: Heartnut (2) | Not in the main orchard (nut tree) | |

Also at Prairie but outside this plan. Berry bushes: Blackberry (various) 3, Hardy Kiwi 4, Sea Berry 11, Rasberry? 1. Perennials: Comfrey 25, Mint (various) 18, Rhubarb 15. Evergreens: Fir (Douglas) 2, Juniper (Eastern Red Ceder) 7, Juniper (Variety) 8, Pine (Austrian Black) 7, Pine (Ponderosa) 18, Pine (White) 2 (owner reports both died), Spruce (Blue) 4. Shade trees: Ash 3, Aspen (Trembling) 15, Maple (Sugar) 10, Maple 1, Oak (various) 8.

## Apple Flowering-Group Research — September 24, 2026

Research covers the apple records that were missing flowering groups plus Harrison, mirrored from the dated notes in the Trees.xlsx F Group cells. Numbered assignments generally follow Orange Pippin; Foxwhelp follows Raintree, its recorded vendor; seed-grown Kerr uses an owner-approved best guess. Flowering groups are not universally standardized, and local bloom overlap still needs checking.

| Apple / inventory record | Trees at Prairie | Workbook FG | Planning bloom / FG | Evidence, limitations, and sources |
|---|---|---|---|---|
| Blushing Delight Urban Columnar<br>Fruit sheet row 6; R5-C10 is open; tree planted outside the orchard (C13-E42-S20) | 1 | Not recorded | Midseason; FG ? | Raintree describes the Urban Apple series as midseason bloomers and recommends different Urban cultivars for one another. No numbered flowering group was verified for Blushing Delight.<br>Raintree: Blushing Delight |
| Campfield Cider Apple<br>Fruit sheet row 9; R5-C1 | 1 | 3 | FG 3 | Orange Pippin explicitly assigns Campfield to flowering group 3. Raintree independently lists bloom group 3 (midseason). Both refer to flowering, not harvest season. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Orange Pippin: Campfield (FG 3)<br>Raintree: Campfield (bloom group 3) |
| Dolgo Crabapple<br>Fruit sheet row 13; R4-C13 | 1 | 2 | FG 2 | Orange Pippin explicitly assigns the named Dolgo cultivar to flowering group 2 and describes early blossom. This assignment assumes these are the named cultivar, not unnamed seedlings. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Orange Pippin: Dolgo |
| Foxwhelp Cider Apple<br>Fruit sheet row 15; R2-C1 | 1 | 3 | FG 3 | The workbook records RainTree as the vendor for this tree, supporting the owner-selected Raintree group 3 (midseason). Raintree identifies its stock as Sonoma/Geneva Foxwhelp (Fauxwhelp), not authenticated English Foxwhelp. Other sources describe different Foxwhelp material (Orange Pippin group 5; RV Roger group B); those classifications are not substituted for the recorded supplier. Confirm local bloom. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Raintree: Fauxwhelp stock (bloom group 3)<br>Orange Pippin: generic Foxwhelp entry (group 5; identity caveat)<br>RV Roger: Foxwhelp (lettered group B) |
| Frequin Rouge Cider Apple<br>Fruit sheet row 17; R2-C13 | 1 | 4 | FG 4* | Orange Pippin assigns Frequin Rouge to group 4, corroborated by Raintree (mid-late). Cummins instead lists group 5 in its five-group system and warns that the precise Frequin Rouge variant in North America is uncertain. Use 4 as the source-attributed planning value, not proof of local bloom or a reason to substitute another Frequin cultivar. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Orange Pippin: Frequin Rouge (FG 4)<br>Raintree: Frequin Rouge (bloom group 4)<br>Cummins: Frequin Rouge (group 5; variant uncertainty)<br>Cummins: five-group bloom scheme |

| Apple / inventory record | Trees at Prairie | Workbook FG | Planning bloom / FG | Evidence, limitations, and sources |
|---|---|---|---|---|
| Harrison<br>Fruit sheet row 26; R4-C12, R5-C11 | 2 | 3 | FG 3* | Orange Pippin assigns Harrison to group 3, matching the existing Harrison workbook entry. The same assignment is used for Harrison Cider Apple, not a different pollen partner. Raintree instead lists group 4 (mid-late); retain this source difference and verify local bloom.<br>Orange Pippin: Harrison (FG 3)<br>Raintree: Harrison (bloom group 4) |
| Kerr Crab Seedling<br>Fruit sheet row 33; R3-C1, R5-C13, R6-C1 | 3 | 2 | FG 2? | BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Orange Pippin: named Kerr cultivar (FG 2 only for the cultivar)<br>Iowa State Extension: apple seedlings do not come true to type |
| Tandgy green Urban Columnar<br>Fruit sheet row 42; R4-C9 is open; tree planted outside the orchard (D13-E0-S20) | 1 | Not recorded | Midseason; FG ? | The workbook spelling "Tandgy green" is treated as Tangy Green Urban Apple (AK-98); confirm the nursery label. Raintree gives midseason bloom guidance for the series and recommends other Urban cultivars. No numbered flowering group was verified.<br>Raintree: Tangy Green |
| Tasty Red Urban Columnar<br>Fruit sheet row 43; R5-C9 is open; tree planted outside the orchard (C13-E46-S20) | 1 | Not recorded | Midseason; FG ? | Raintree explicitly describes Urban Apple varieties as midseason bloomers and recommends them for one another. No numbered flowering group was verified for Tasty Red.<br>Raintree: Tasty Red |

## Pollination Layout Check

Counts use the same screening as the partner lists below. Self-fertile trees are not counted as needing cross-pollination. Unidentified trees and trees without bloom or compatibility data have no screened partner yet; proximity alone is not counted as evidence.

| Fruit type | Rows | Trees | Need cross-pollination | With a screened candidate ≤ 50 ft | With 2 distinct candidates ≤ 50 ft | Triploid/disputed with 2 mutual donors ≤ 50 ft | No screened partner yet | Adjacent same-cultivar pairs |
|---|---|---|---|---|---|---|---|---|
| Apples | R1–R6 | 51 | 45 | 39 of 45 | 39 of 45 | 5 of 5 | 6 | 0 |
| Pears | R7–R8 | 24 | 21 | 16 of 21 | 12 of 21 | — | 5 | 1 |
| Cherries | R9–R10 | 15 | 11 | 4 of 11 | 0 of 11 | — | 7 | 0 |
| Peaches | R11–R12 | 8 | 2 | 0 of 2 | 0 of 2 | — | 2 | 0 |
| Apricots | R11–R12 | 5 | 0 | 0 of 0 | 0 of 0 | — | 0 | 0 |
| Plums | R11–R12 | 3 | 0 | 0 of 0 | 0 of 0 | — | 0 | 3 |

## Pollination Design Notes

All 106 living orchard fruit trees and 1 planted rootstock are placed once, honoring the inventory after owner planting updates. The 3 Urban apples are planted outside the orchard, so they are not counted as orchard pollen partners. Each fruit type has its own rows except R11–R12, which the apricots, plums, and peaches share. Open positions (75 in all) are at the east ends, between the blocks in the shared rows, and across R13–R15, which are held for trees you buy or graft later, and at R4-C9, R5-C9, R5-C10, R8-C1, R9-C2, R11-C10, R11-C12, R12-C11 after the owner's planting updates; they are not additional trees.

Row layouts: apples in 5 rows (13 + 13 + 12 + 11 + 3) plus reserved R1 (13 spots); pears in 2 rows (13 + 11); cherries in 2 rows (7 + 8); apricots at the west end (3 + 2), plums at C6 in R11 and C5–C6 in R12 (1 + 2), and peaches at the east end (4 + 4) share R11–R12; R13–R15 are open for trees you buy or graft later. Single-type blocks start at the west end, and a type's shorter row sits against its fuller rows to keep partners close.

Within each type, positions were chosen by a search that uses the same screening rules shown on this page. It favors two distinct candidates within 50 ft, two mutually screened diploid donors for triploid and disputed apples, and separation of copies of the same cultivar. It is a screened arrangement, not a guarantee of the best possible one.

Apple candidates use the workbook pollenizer lists and same/adjacent flowering groups, supplemented by the linked bloom research. The three Urban apples also use Raintree's within-series recommendation. Known triploids, disputed Liberty, unidentified trees, and unverified pollen donors are excluded as suppliers. Fuji/Aztec/Auvil, Honeycrisp/Super Red, and Bartlett/Red Bartlett count as shared cultivar identities.

Pear bloom charts are a screening tool, not proof of cross-compatibility. Asian pears are grouped without assuming European pears will pollinate them, and perry pears with different bloom windows are not treated as interchangeable.

Sweet cherries use compatibility groups plus same/adjacent bloom periods. Lapins is the only identified sweet cherry within one bloom period of both Black Tartarian and Bing, so the sweet cherries are grouped around it. Stella is not relied on for Bing. Montmorency is a self-fertile tart cherry, and the Nanking bushes are kept together and are not sweet-cherry donors.

J.H. Hale was the only peach needing a partner, and it died (owner report, Oct 17). The remaining identified peaches are self-fertile; unidentified peaches are not counted as donors.

11 unidentified trees (6 apples, 3 sweet cherries, 2 peaches) are placed among identified trees of their type but are not counted as donors until identified.

20 trees that are not self-fertile still have no screened partner, mostly unidentified trees and trees missing bloom or compatibility data. They are marked unconfirmed below; proximity alone is not counted as pollination evidence.

| Row | Fruit type | Trees | Open | Placement notes |
|---|---|---|---|---|
| R1 | Apple (reserved) | 0 (13 reserved) | 0 | Reserved for 13 grafted rootstock apples; 14 rootstocks remain available (2 EMLA 7, 11 M-111, 1 unknown), after the M-111 planted at R6-C3. Choose scion varieties whose bloom overlaps the apples in the next rows. |
| R2 | Apple | 13 | 0 | Planning bloom: FG 3, FG 4, FG 4*. |
| R3 | Apple | 13 | 0 | Planning bloom: FG 2?, FG 3, FG 4, FG ?. |

| Row | Fruit type | Trees | Open | Placement notes |
|---|---|---|---|---|
| R4 | Apple | 12 | 1 | Planning bloom: FG 1, FG 2, FG 3, FG 3*, FG 4, FG ?. R4-C9: Tandgy green Urban Columnar (tag AL-UrbGrn, inventory row 64) not planted in the orchard. This living tree is planted outside the orchard; SiteRef D13-E0-S20. Leave this spot open; the plan is a new Haralson (apple) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock M.7, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan. |
| R5 | Apple | 11 | 2 | Planning bloom: FG 1, FG 2, FG 2?, FG 3, FG 3*. R5-C9: Tasty Red Urban Columnar (tag AL-UrbR, inventory row 65) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E46-S20. Leave this spot open; the plan is a new Williams' Pride (apple) in 2027. It is ordered and has not arrived: Cummins Nursery order 039970, 2026-10-27, ships to Idaho. Rootstock B.118, grade 1, Spring 2027. Described in the planting plan. R5-C10: Blushing Delight Urban Columnar (tag AL-UrbGld, inventory row 63) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E42-S20. Leave this spot open; the plan is a new Priscilla (apple) in 2027. Described in the planting plan. |
| R6 | Apple | 3 | 10 | Planning bloom: FG 2?, FG 3. M-111 apple rootstock planted at R6-C3. No scion cultivar is recorded; not counted as a fruit tree or pollen partner until grafting and cultivar identification are recorded. |
| R7 | Pear | 13 | 0 | European/perry bloom windows and the Asian group checked separately. |
| R8 | Pear | 11 | 2 | European/perry bloom windows and the Asian group checked separately. R8-C1: Pineapple (tag PR-Pineap, inventory row 108) reported dead October 11, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Moonglow, stand-in for Harrow Delight (pear) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock OHxF 97, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan. |
| R9 | Cherry | 7 | 6 | Sweet cherries grouped around Lapins; tart and Nanking kept out of the sweet-cherry network. R9-C2: Royalton Sweet (tag CR-Royalton, inventory row 80) reported dead October 17, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Black Pearl (cherry) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order XFEEV5ANY, 2026-10-28, ships to Idaho. Rootstock Mazzard, grade Hand Select, Spring 2027. Described in the planting plan. |
| R10 | Cherry | 8 | 5 | Sweet cherries grouped around Lapins; tart and Nanking kept out of the sweet-cherry network. |
| R11 | Apricot, plum, and peach (shared) | 8 | 5 | Apricots C1–C3 at the west end, plum C6, and peaches C8, C9, C11, C13 at the east end. Compact block of self-fertile apricots. Italian plums, self-fertile. All identified peaches are self-fertile (J.H. Hale died). |
| R12 | Apricot, plum, and peach (shared) | 8 | 5 | Apricots C1–C2 at the west end, plums C5–C6, and peaches C9, C10, C12, C13 at the east end. Compact block of self-fertile apricots. Italian plums, self-fertile. All identified peaches are self-fertile (J.H. Hale died). |
| R13 | Future trees | 0 | 13 | Held open for trees you buy or graft onto your rootstock later. Apples grafted here would be 140 ft from the nearest apples (R6), so plant at least two cultivars with overlapping bloom together. |
| R14 | Future trees | 0 | 13 | Held open for trees you buy or graft onto your rootstock later. Apples grafted here would be 160 ft from the nearest apples (R6), so plant at least two cultivars with overlapping bloom together. |
| R15 | Future trees | 0 | 13 | Held open for trees you buy or graft onto your rootstock later. Apples grafted here would be 180 ft from the nearest apples (R6), so plant at least two cultivars with overlapping bloom together. |

## Priority Pollination Pairings

Trees that are not self-fertile, plus triploid or disputed apples. Candidates are screened by fruit type, bloom timing, and recorded compatibility; distances are between tree centers.

| Tree needing pollen | Position | Nearby candidates | Evidence / limitation |
|---|---|---|---|
| Foxwhelp Cider Apple | R2-C1 | Braeburn (R2-C2, 18.0 ft) - Bloom-group candidate; Cortland (R3-C2, 21.9 ft) - Workbook-listed + bloom group | The workbook records RainTree as the vendor for this tree, supporting the owner-selected Raintree group 3 (midseason). Raintree identifies its stock as Sonoma/Geneva Foxwhelp (Fauxwhelp), not authenticated English Foxwhelp. Other sources describe different Foxwhelp material (Orange Pippin group 5; RV Roger group B); those classifications are not substituted for the recorded supplier. Confirm local bloom. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. |
| Jonagold | R2-C3 | Braeburn (R2-C2, 18.0 ft) - Bloom-group candidate; Fuji (R2-C4, 18.0 ft) - Workbook-listed + bloom group | Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required. |
| Fuji | R2-C4 | Honeycrisp (R2-C5, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R3-C5, 21.9 ft) - Workbook-listed + bloom group | Confirm overlapping bloom locally before relying on these candidates. |
| Honeycrisp | R2-C5 | Cortland (R2-C6, 18.0 ft) - Bloom-group candidate; Fuji (R2-C4, 18.0 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Cortland | R2-C6 | Honeycrisp (R2-C5, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R2-C7, 18.0 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Pink Lady (Cripps Pink) | R2-C7 | Cortland (R2-C6, 18.0 ft) - Bloom-group candidate; Honeycrisp (R3-C8, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Jonagold | R2-C8 | Gala (R2-C9, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R2-C7, 18.0 ft) - Workbook-listed + bloom group | Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required. |
| Arkansas Black Spur (FT015) | R2-C10 | Cox's Orange Pippin (R2-C11, 18.0 ft) - Bloom-group candidate; Gala (R2-C9, 18.0 ft) - Bloom-group candidate | Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required. |
| Franklin Cider | R2-C12 | Cox's Orange Pippin (R2-C11, 18.0 ft) - Workbook-listed + bloom group; Newtown Pippen (R3-C13, 21.9 ft) - Workbook-listed + bloom group | Confirm overlapping bloom locally before relying on these candidates. |
| Frequin Rouge Cider Apple | R2-C13 | Franklin Cider (R2-C12, 18.0 ft) - Bloom-group candidate; Newtown Pippen (R3-C13, 21.9 ft) - Workbook-listed + bloom group | Orange Pippin assigns Frequin Rouge to group 4, corroborated by Raintree (mid-late). Cummins instead lists group 5 in its five-group system and warns that the precise Frequin Rouge variant in North America is uncertain. Use 4 as the source-attributed planning value, not proof of local bloom or a reason to substitute another Frequin cultivar. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. |
| Kerr Crab Seedling | R3-C1 | Cortland (R3-C2, 18.0 ft) - Bloom-group candidate; provisional flowering group; Cox's Orange Pippin (R4-C1, 21.9 ft) - Bloom-group candidate; provisional flowering group | BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. |
| Cortland | R3-C2 | Kingston Black (R3-C3, 18.0 ft) - Workbook-listed + bloom group; Braeburn (R2-C2, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Kingston Black | R3-C3 | Cortland (R3-C2, 18.0 ft) - Bloom-group candidate; Braeburn (R2-C2, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Mutsu Crispin | R3-C4 | Kingston Black (R3-C3, 18.0 ft) - Bloom-group candidate; Pink Lady (Cripps Pink) (R3-C5, 18.0 ft) - Bloom-group candidate | Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required. |
| Pink Lady (Cripps Pink) | R3-C5 | Aztec Fuji (R3-C6, 18.0 ft) - Bloom-group candidate; Golden Delicious (R4-C4, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Aztec Fuji | R3-C6 | Pink Lady (Cripps Pink) (R3-C5, 18.0 ft) - Bloom-group candidate; Cortland (R2-C6, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Unidentified apple | R3-C7 | Unconfirmed — no screened partner in the available data. | Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. |
| Honeycrisp | R3-C8 | Aztec Fuji (R3-C9, 18.0 ft) - Bloom-group candidate; Pink Lady (Cripps Pink) (R2-C7, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Aztec Fuji | R3-C9 | GoldRush (R3-C10, 18.0 ft) - Bloom-group candidate; Honeycrisp (R3-C8, 18.0 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| GoldRush | R3-C10 | Red Delicious (R3-C11, 18.0 ft) - Bloom-group candidate; Aztec Fuji (R3-C9, 18.0 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Red Delicious | R3-C11 | GoldRush (R3-C10, 18.0 ft) - Bloom-group candidate; Cox's Orange Pippin (R2-C11, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |

| Tree needing pollen | Position | Nearby candidates | Evidence / limitation |
|---|---|---|---|
| Unidentified apple | R3-C12 | Unconfirmed — no screened partner in the available data. | Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. |
| Unidentified apple | R4-C2 | Unconfirmed — no screened partner in the available data. | Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. |
| Honeycrisp (Super Red) | R4-C3 | Golden Delicious (R4-C4, 18.0 ft) - Bloom-group candidate; Hudson's Golden Gem (R5-C3, 21.9 ft) - Workbook-listed + bloom group | Confirm overlapping bloom locally before relying on these candidates. |
| Golden Delicious | R4-C4 | Honeycrisp (Super Red) (R4-C3, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R3-C5, 21.9 ft) - Workbook-listed + bloom group | Confirm overlapping bloom locally before relying on these candidates. |
| Unidentified apple | R4-C5 | Unconfirmed — no screened partner in the available data. | Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. |
| Zestar | R4-C6 | Wickson Crab (R4-C7, 18.0 ft) - Bloom-group candidate; Macintosh (R5-C7, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Wickson Crab | R4-C7 | Zestar (R4-C6, 18.0 ft) - Bloom-group candidate; Macintosh (R5-C7, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Unidentified apple | R4-C8 | Unconfirmed — no screened partner in the available data. | Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. |
| Unidentified apple | R4-C10 | Unconfirmed — no screened partner in the available data. | Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners. |
| Auvil Early Fuji | R4-C11 | Harrison (R4-C12, 18.0 ft) - Bloom-group candidate; CrimsonCrisp (R5-C12, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Harrison | R4-C12 | Auvil Early Fuji (R4-C11, 18.0 ft) - Bloom-group candidate; CrimsonCrisp (R5-C12, 21.9 ft) - Bloom-group candidate | Orange Pippin assigns Harrison to group 3, matching the existing Harrison workbook entry. The same assignment is used for Harrison Cider Apple, not a different pollen partner. Raintree instead lists group 4 (mid-late); retain this source difference and verify local bloom. |
| Dolgo Crabapple | R4-C13 | Harrison (R4-C12, 18.0 ft) - Workbook-listed + bloom group; Newtown Pippen (R3-C13, 21.9 ft) - Workbook-listed + bloom group | Orange Pippin explicitly assigns the named Dolgo cultivar to flowering group 2 and describes early blossom. This assignment assumes these are the named cultivar, not unnamed seedlings. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. |
| Campfield Cider Apple | R5-C1 | Idared (R5-C2, 18.0 ft) - Bloom-group candidate; Cox's Orange Pippin (R4-C1, 21.9 ft) - Workbook-listed + bloom group | Orange Pippin explicitly assigns Campfield to flowering group 3. Raintree independently lists bloom group 3 (midseason). Both refer to flowering, not harvest season. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. |
| Idared | R5-C2 | Hudson's Golden Gem (R5-C3, 18.0 ft) - Workbook-listed + bloom group; Cox's Orange Pippin (R4-C1, 21.9 ft) - Workbook-listed + bloom group | Confirm overlapping bloom locally before relying on these candidates. |
| Hudson's Golden Gem | R5-C3 | Idared (R5-C2, 18.0 ft) - Bloom-group candidate; Granny Smith (R6-C2, 21.9 ft) - Workbook-listed + bloom group | Confirm overlapping bloom locally before relying on these candidates. |
| Liberty | R5-C4 | Hudson's Golden Gem (R5-C3, 18.0 ft) - Bloom-group candidate; Sweet Sixteen (R5-C5, 18.0 ft) - Bloom-group candidate | Liberty ploidy/donor status is disputed. Excluded as a donor; given two diploid candidates. |
| Sweet Sixteen | R5-C5 | Cortland (R5-C6, 18.0 ft) - Bloom-group candidate; Golden Delicious (R4-C4, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Cortland | R5-C6 | Macintosh (R5-C7, 18.0 ft) - Bloom-group candidate; Sweet Sixteen (R5-C5, 18.0 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Macintosh | R5-C7 | Cortland (R5-C6, 18.0 ft) - Bloom-group candidate; Zestar (R5-C8, 18.0 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Zestar | R5-C8 | Macintosh (R5-C7, 18.0 ft) - Bloom-group candidate; Wickson Crab (R4-C7, 21.9 ft) - Bloom-group candidate | Confirm overlapping bloom locally before relying on these candidates. |
| Harrison | R5-C11 | CrimsonCrisp (R5-C12, 18.0 ft) - Bloom-group candidate; Auvil Early Fuji (R4-C11, 21.9 ft) - Bloom-group candidate | Orange Pippin assigns Harrison to group 3, matching the existing Harrison workbook entry. The same assignment is used for Harrison Cider Apple, not a different pollen partner. Raintree instead lists group 4 (mid-late); retain this source difference and verify local bloom. |
| CrimsonCrisp | R5-C12 | Harrison (R5-C11, 18.0 ft) - Workbook-listed + bloom group; Auvil Early Fuji (R4-C11, 21.9 ft) - Workbook-listed + bloom group | Confirm overlapping bloom locally before relying on these candidates. |
| Kerr Crab Seedling | R5-C13 | CrimsonCrisp (R5-C12, 18.0 ft) - Bloom-group candidate; provisional flowering group; Harrison (R4-C12, 21.9 ft) - Bloom-group candidate; provisional flowering group | BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. |

| Tree needing pollen | Position | Nearby candidates | Evidence / limitation |
|---|---|---|---|
| Kerr Crab Seedling | R6-C1 | Granny Smith (R6-C2, 18.0 ft) - Bloom-group candidate; provisional flowering group; Idared (R5-C2, 21.9 ft) - Bloom-group candidate; provisional flowering group | BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it. |
| Yellow Huffcapp European Perry Pear | R7-C1 | Unconfirmed — no screened partner in the available data. | Yellow Huffcap bloom and compatibility need confirmation; do not assume all perry pears overlap. |
| Shinseiki - Asian | R7-C2 | Unconfirmed — no screened partner in the available data. | Shinseiki is grouped near the other Asian pears; a named compatible pair remains unconfirmed. |
| Olympic Giant - Asian | R7-C3 | Unconfirmed — no screened partner in the available data. | Confirm Asian-pear nursery identity and bloom. The two Giant selections are not counted as independent partners. |
| Red Bartlett | R7-C4 | Anjou (R7-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Hendre Huffcapp European Perry Pear (R8-C4, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm overlapping bloom locally before relying on these candidates. |
| Anjou | R7-C5 | Comice (R7-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Red Bartlett (R7-C4, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm overlapping bloom locally before relying on these candidates. |
| Barnett European Perry Pear | R7-C7 | Unconfirmed — no screened partner in the available data. | Confirm whether Barnett is Barnet. Barnet is very late in the perry chart; no partner is assumed. |
| SummerCrisp | R7-C8 | Parker (R7-C9, 18.0 ft) - Nursery-recommended Parker for Summercrisp; confirm local bloom | Confirm overlapping bloom locally before relying on these candidates. |
| Parker | R7-C9 | Unconfirmed — no screened partner in the available data. | Parker is recommended for Summercrisp; the reverse pairing was not verified. |
| SummerCrisp | R7-C10 | Parker (R7-C9, 18.0 ft) - Nursery-recommended Parker for Summercrisp; confirm local bloom | Confirm overlapping bloom locally before relying on these candidates. |
| Keiffer | R7-C12 | Comice (R7-C11, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Butt European Perry Pear (R7-C13, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm overlapping bloom locally before relying on these candidates. |
| Butt European Perry Pear | R7-C13 | Keiffer (R7-C12, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Comice (R8-C12, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility | Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility. |
| Hosui - Asian | R8-C2 | Starking Hardy Giant - Asian (R8-C3, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm overlapping bloom locally before relying on these candidates. |
| Starking Hardy Giant - Asian | R8-C3 | Hosui - Asian (R8-C2, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm Asian-pear nursery identity and bloom. The two Giant selections are not counted as independent partners. |
| Hendre Huffcapp European Perry Pear | R8-C4 | Bartlett (R8-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Anjou (R7-C5, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility | Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility. |
| Bartlett | R8-C5 | Hendre Huffcapp European Perry Pear (R8-C4, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Moonglow (R8-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm overlapping bloom locally before relying on these candidates. |
| Moonglow | R8-C6 | Anjou (R8-C7, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Bartlett (R8-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm overlapping bloom locally before relying on these candidates. |
| Anjou | R8-C7 | Hendre Huffcapp European Perry Pear (R8-C8, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Moonglow (R8-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm overlapping bloom locally before relying on these candidates. |
| Hendre Huffcapp European Perry Pear | R8-C8 | Anjou (R8-C7, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Starking Delicious (R8-C9, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility. |
| Starking Delicious | R8-C9 | Anjou (R8-C10, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Hendre Huffcapp European Perry Pear (R8-C8, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm overlapping bloom locally before relying on these candidates. |
| Anjou | R8-C10 | Starking Delicious (R8-C9, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Bartlett (R8-C11, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm overlapping bloom locally before relying on these candidates. |
| Bartlett | R8-C11 | Comice (R8-C12, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Anjou (R8-C10, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | Confirm overlapping bloom locally before relying on these candidates. |
| Bing | R9-C1 | Lapins (R10-C2, 33.6 ft) - Cherry compatibility + bloom chart; confirm local overlap | Treat Bing as cross-pollination dependent. Stella is conservatively excluded as its donor. |
| Bing | R9-C3 | Lapins (R10-C2, 21.9 ft) - Cherry compatibility + bloom chart; confirm local overlap | Treat Bing as cross-pollination dependent. Stella is conservatively excluded as its donor. |
| Nanking (bush) | R9-C5 | Unconfirmed — no screened partner in the available data. | Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines. |
| Unidentified sweet cherry | R9-C6 | Unconfirmed — no screened partner in the available data. | Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified. |
| Hardy Giant Antque Sweet | R9-C7 | Unconfirmed — no screened partner in the available data. | Sweet-cherry donor compatibility was not verified; do not rely on this tree to supply other varieties. |
| Nanking (bush) | R9-C8 | Unconfirmed — no screened partner in the available data. | Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines. |
| Black Tartarian | R10-C1 | Lapins (R10-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap | Black Tartarian blooms very early; Lapins is its only identified donor in an adjacent bloom period. Two of the four purchased Black Tartarians were not found and may be among the unidentified sweet cherries. |

| Tree needing pollen | Position | Nearby candidates | Evidence / limitation |
|---|---|---|---|
| Black Tartarian | R10-C3 | Lapins (R10-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap | Black Tartarian blooms very early; Lapins is its only identified donor in an adjacent bloom period. Two of the four purchased Black Tartarians were not found and may be among the unidentified sweet cherries. |
| Unidentified sweet cherry | R10-C4 | Unconfirmed — no screened partner in the available data. | Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified. |
| Nanking (bush) | R10-C6 | Unconfirmed — no screened partner in the available data. | Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines. |
| Unidentified sweet cherry | R10-C7 | Unconfirmed — no screened partner in the available data. | Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified. |
| Unidentified peach | R12-C10 | Unconfirmed — no screened partner in the available data. | Unidentified peach, possibly Contender, Elberta, Reliance, or Veteran. Not counted as a pollen donor until identified. |
| Unidentified peach | R12-C12 | Unconfirmed — no screened partner in the available data. | Unidentified peach, possibly Contender, Elberta, Reliance, or Veteran. Not counted as a pollen donor until identified. |

## Full Tree Inventory by Row

Living orchard trees and records of vacated planting positions, south (R1) to north and west (C1) to east. Dead and outside-orchard trees retain their original planned slots as history; those slots are open, not their current locations. They are excluded from orchard counts and pollen partners. The planted M-111 rootstock occupies R6-C3 but is not counted as a fruit tree or pollen partner without a recorded scion cultivar. The Pineapple's open slot moved from R8-C3 to R8-C1 when the owner found the Asian pear (PR-Asian) in R8-C3. Original workbook fertility values are retained.
Spot SiteRef identifies the listed planned or vacated orchard position, not an outside-orchard tree's current location. It is generated from the shared grid coordinates and is separate from the metal tag and individual Plant ID.

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R2-C1 | A12-E47-S30 | **FOX** | **Foxwhelp Cider Apple**<br>Tag AL-Foxwhelp; inventory name: Foxwhelp Cider<br>Late; crimson-striped; bittersharp cider apple<br>Cultivar profile | Apple | Semi Dwarf | FG 3 | Not recorded | Planted<br>The workbook records RainTree as the vendor for this tree, supporting the owner-selected Raintree group 3 (midseason). Raintree identifies its stock as Sonoma/Geneva Foxwhelp (Fauxwhelp), not authenticated English Foxwhelp. Other sources describe different Foxwhelp material (Orange Pippin group 5; RV Roger group B); those classifications are not substituted for the recorded supplier. Confirm local bloom. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Braeburn (R2-C2, 18.0 ft) - Bloom-group candidate; Cortland (R3-C2, 21.9 ft) - Workbook-listed + bloom group<br>Bloom research and sources | 39 (15) |
| R2-C2 | B12-E15-S30 | **BB** | **Braeburn**<br>Tag AL-Brae Dwf; inventory name: Braeburn Dwarf<br>Late; red-orange streaks; crisp, tangy-sweet<br>Cultivar profile | Apple | Not recorded (tag says Dwarf) | FG 4 | Self-fertile | Self-fertile Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>Cortland (R3-C2, 21.9 ft) - Workbook-listed + bloom group; Kingston Black (R3-C3, 21.9 ft) - Bloom-group candidate<br>Inventory match: 1 of 2 purchased Braeburns found. The tag says Dwarf, so it is matched to the Lowes tree with no recorded rootstock (row 8) rather than the M-111 tree (row 7). | 27 (8, 7) |
| R2-C3 | B12-E33-S30 | **JG** | **Jonagold**<br>Tag AL-Jon G; inventory name: Jonagold<br>Late; large red-over-gold; honeyed sweet-tart; triploid<br>Cultivar profile | Apple | Semi Dwarf | FG 4 | Not self-fertile | Triploid Planted<br>Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required.<br>Braeburn (R2-C2, 18.0 ft) - Bloom-group candidate; Fuji (R2-C4, 18.0 ft) - Workbook-listed + bloom group | 53 (32) |
| R2-C4 | C12-E1-S30 | **FJ** | **Fuji**<br>Tag AL-Fuji; inventory name: Fuji<br>Very late; pink-red stripes; sweet, crisp, long keeper<br>Cultivar profile | Apple | Semi Dwarf | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Honeycrisp (R2-C5, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R3-C5, 21.9 ft) - Workbook-listed + bloom group<br>Inventory match: Same tag text (AL-Fuji) as the Early Fuji; check the variety label when planting. | 42 (18) |
| R2-C5 | C12-E19-S30 | **HC** | **Honeycrisp**<br>Tag AL-Hon Cris; inventory name: Honeycrisp<br>Mid; red-orange flush; very crisp, sweet<br>Cultivar profile | Apple | M-111 | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Cortland (R2-C6, 18.0 ft) - Bloom-group candidate; Fuji (R2-C4, 18.0 ft) - Bloom-group candidate<br>Inventory match: 2 of 3 purchased found. | 49 (28) |
| R2-C6 | C12-E37-S30 | **CL** | **Cortland**<br>Tag AL-Cort/ AL-Cortland; inventory name: Cortland<br>Mid; red-streaked; white flesh slow to brown<br>Cultivar profile | Apple | M-111 | FG 3 | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Honeycrisp (R2-C5, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R2-C7, 18.0 ft) - Bloom-group candidate<br>Inventory match: 3 of 5 purchased found. | 29 (10) |
| R2-C7 | D12-E5-S30 | **PL** | **Pink Lady (Cripps Pink)**<br>Tag AL-Pink/ AL-Pink Lady; inventory name: Pink Lady<br>Very late; pink blush on yellow; crisp, sweet-tart<br>Cultivar profile | Apple | Not recorded | FG 3 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Cortland (R2-C6, 18.0 ft) - Bloom-group candidate; Honeycrisp (R3-C8, 21.9 ft) - Bloom-group candidate | 59 (39) |
| R2-C8 | D12-E23-S30 | **JG** | **Jonagold**<br>Tag AL-Jon G; inventory name: Jonagold<br>Late; large red-over-gold; honeyed sweet-tart; triploid<br>Cultivar profile | Apple | Semi Dwarf | FG 4 | Not self-fertile | Triploid Planted<br>Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required.<br>Gala (R2-C9, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R2-C7, 18.0 ft) - Workbook-listed + bloom group | 53 (32) |
| R2-C9 | D12-E41-S30 | **GL** | **Gala**<br>Tag AL-Gala; inventory name: Gala<br>Early; orange-red streaks; mild, sweet<br>Cultivar profile | Apple | M-111 | FG 4 | Self-fertile | Self-fertile Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>Aztec Fuji (R3-C9, 21.9 ft) - Workbook-listed + bloom group; GoldRush (R3-C10, 21.9 ft) - Workbook-listed + bloom group<br>Inventory match: 1 of 2 purchased found. | 45 (19, 20) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R2-C10 | E12-E9-S30 | **AB** | **Arkansas Black Spur (FT015)**<br>Tag AL-Ark Blk; inventory name: Arkansas Black<br>Very late; dark red-black; tart, firm, long keeper<br>Cultivar profile | Apple | M-111 | FG 3 | Not self-fertile | Triploid Planted<br>Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required.<br>Cox's Orange Pippin (R2-C11, 18.0 ft) - Bloom-group candidate; Gala (R2-C9, 18.0 ft) - Bloom-group candidate | 26 (2) |
| R2-C11 | E12-E27-S30 | **CO** | **Cox's Orange Pippin**<br>Tag AL-O Pip; inventory name: Cox Orange Pippin<br>Mid; orange-red flush; aromatic, sweet-tart; fussy<br>Cultivar profile | Apple | M-7 | FG 3 | Self-fertile | Self-fertile Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>Franklin Cider (R2-C12, 18.0 ft) - Bloom-group candidate; Red Delicious (R3-C11, 21.9 ft) - Bloom-group candidate | 30 (11) |
| R2-C12 | E12-E45-S30 | **FC** | **Franklin Cider**<br>Tag AL-Frank C; inventory name: Franklin Cider<br>Late; russeted green-yellow; bittersharp cider apple<br>Cultivar profile | Apple | Semi Dwarf | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Cox's Orange Pippin (R2-C11, 18.0 ft) - Workbook-listed + bloom group; Newtown Pippen (R3-C13, 21.9 ft) - Workbook-listed + bloom group | 40 (16) |
| R2-C13 | F12-E13-S30 | **FRQ** | **Frequin Rouge Cider Apple**<br>Tag AL-Frequin; inventory name: Frequin Rouge<br>Mid; small red-striped; bittersweet French cider apple<br>Cultivar profile | Apple | Semi Dwarf | FG 4* | Not recorded | Planted<br>Orange Pippin assigns Frequin Rouge to group 4, corroborated by Raintree (mid-late). Cummins instead lists group 5 in its five-group system and warns that the precise Frequin Rouge variant in North America is uncertain. Use 4 as the source-attributed planning value, not proof of local bloom or a reason to substitute another Frequin cultivar. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Franklin Cider (R2-C12, 18.0 ft) - Bloom-group candidate; Newtown Pippen (R3-C13, 21.9 ft) - Workbook-listed + bloom group<br>Bloom research and sources | 41 (17) |
| R3-C1 | A12-E38-S10 | **KER** | **Kerr Crab Seedling**<br>Tag AL-Kerr Crab; inventory name: Crabapple, Kerr<br>Mid; crimson-maroon crab; crisp, tart-sweet; keeps<br>Cultivar profile | Apple | Not recorded | FG 2? | Not recorded | Planted<br>BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Cortland (R3-C2, 18.0 ft) - Bloom-group candidate; provisional flowering group; Cox's Orange Pippin (R4-C1, 21.9 ft) - Bloom-group candidate; provisional flowering group<br>Inventory match: 3 found, but 2 were purchased. All are seed-grown, so each may bloom differently.<br>Bloom research and sources | 32 (33) |
| R3-C2 | B12-E6-S10 | **CL** | **Cortland**<br>Tag AL-Cort/ AL-Cortland; inventory name: Cortland<br>Mid; red-streaked; white flesh slow to brown<br>Cultivar profile | Apple | M-111 | FG 3 | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Kingston Black (R3-C3, 18.0 ft) - Workbook-listed + bloom group; Braeburn (R2-C2, 21.9 ft) - Bloom-group candidate<br>Inventory match: 3 of 5 purchased found. | 29 (10) |
| R3-C3 | B12-E24-S10 | **KB** | **Kingston Black**<br>Tag AL-G Orch-7; inventory name: Kingston Black<br>Late; dark red, russeted; bittersharp cider apple<br>Cultivar profile | Apple | M-7 | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Cortland (R3-C2, 18.0 ft) - Bloom-group candidate; Braeburn (R2-C2, 21.9 ft) - Bloom-group candidate<br>Inventory match: Tag AL-G Orch-7 (Grandpa's Orchard, M-7) matches the workbook. | 54 (34) |
| R3-C4 | B12-E42-S10 | **MC** | **Mutsu Crispin**<br>Tag AL-Crisp; inventory name: Crispin<br>Late; big yellow-green; crisp, honeyed; triploid<br>Cultivar profile | Apple | M-111 | FG 3 | Not self-fertile | Triploid Planted<br>Triploid: not a pollen donor. Two distinct, mutually screened diploid candidates required.<br>Kingston Black (R3-C3, 18.0 ft) - Bloom-group candidate; Pink Lady (Cripps Pink) (R3-C5, 18.0 ft) - Bloom-group candidate | 35 (37) |
| R3-C5 | C12-E10-S10 | **PL** | **Pink Lady (Cripps Pink)**<br>Tag AL-Pink/ AL-Pink Lady; inventory name: Pink Lady<br>Very late; pink blush on yellow; crisp, sweet-tart<br>Cultivar profile | Apple | Not recorded | FG 3 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Aztec Fuji (R3-C6, 18.0 ft) - Bloom-group candidate; Golden Delicious (R4-C4, 21.9 ft) - Bloom-group candidate | 59 (39) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R3-C6 | C12-E28-S10 | **FA** | **Aztec Fuji**<br>Tag AL-Az Fuji; inventory name: Fuji Aztec<br>Very late; deep pink-red; very sweet, long keeper<br>Cultivar profile | Apple | Semi Dwarf or Dwarf | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Pink Lady (Cripps Pink) (R3-C5, 18.0 ft) - Bloom-group candidate; Cortland (R2-C6, 21.9 ft) - Bloom-group candidate<br>Inventory match: 2 of 3 purchased found (row 4: 1 semi-dwarf; row 5: 2 dwarf); the inventory does not say which. | 43 (4, 5) |
| R3-C7 | C12-E46-S10 | **UNK** | **Unidentified apple**<br>Tag AL-UnKn; inventory name: Unknown | Apple | Not recorded | FG ? | Not recorded | Unidentified Planted<br>Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag AL-UnKn. Owner (Oct 16, 2026): every purchased apple arrived, so a purchase with no tagged tree is presumed to have died in Boise, unrecorded. These 6 are among the 11 purchased apples with no tagged tree: Braeburn 1, Cortland 2, Dolgo 1 (long shot), Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Identify them at bloom or harvest using the possibility list.<br>Unidentified apple possibility list | 62 |
| R3-C8 | D12-E14-S10 | **HC** | **Honeycrisp**<br>Tag AL-Hon Cris; inventory name: Honeycrisp<br>Mid; red-orange flush; very crisp, sweet<br>Cultivar profile | Apple | M-111 | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Aztec Fuji (R3-C9, 18.0 ft) - Bloom-group candidate; Pink Lady (Cripps Pink) (R2-C7, 21.9 ft) - Bloom-group candidate<br>Inventory match: 2 of 3 purchased found. | 49 (28) |
| R3-C9 | D12-E32-S10 | **FA** | **Aztec Fuji**<br>Tag AL-Az Fuji; inventory name: Fuji Aztec<br>Very late; deep pink-red; very sweet, long keeper<br>Cultivar profile | Apple | Semi Dwarf or Dwarf | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>GoldRush (R3-C10, 18.0 ft) - Bloom-group candidate; Honeycrisp (R3-C8, 18.0 ft) - Bloom-group candidate<br>Inventory match: 2 of 3 purchased found (row 4: 1 semi-dwarf; row 5: 2 dwarf); the inventory does not say which. | 43 (4, 5) |
| R3-C10 | E12-E0-S10 | **GO** | **GoldRush**<br>Tag AL-Gld Rsh; inventory name: Goldrush<br>Very late; yellow, russet; tart-sweet, keeps 6 months<br>Cultivar profile | Apple | B-118 | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Red Delicious (R3-C11, 18.0 ft) - Bloom-group candidate; Aztec Fuji (R3-C9, 18.0 ft) - Bloom-group candidate | 46 (24) |
| R3-C11 | E12-E18-S10 | **RD** | **Red Delicious**<br>Tag AL-Red Del; inventory name: Delicious, Red<br>Late; tall dark red; sweet, mild, crisp; five bumps<br>Cultivar profile | Apple | M-111 | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>GoldRush (R3-C10, 18.0 ft) - Bloom-group candidate; Cox's Orange Pippin (R2-C11, 21.9 ft) - Bloom-group candidate | 37 (40) |
| R3-C12 | E12-E36-S10 | **UNK** | **Unidentified apple**<br>Tag AL-UnKn; inventory name: Unknown | Apple | Not recorded | FG ? | Not recorded | Unidentified Planted<br>Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag AL-UnKn. Owner (Oct 16, 2026): every purchased apple arrived, so a purchase with no tagged tree is presumed to have died in Boise, unrecorded. These 6 are among the 11 purchased apples with no tagged tree: Braeburn 1, Cortland 2, Dolgo 1 (long shot), Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Identify them at bloom or harvest using the possibility list.<br>Unidentified apple possibility list | 62 |
| R3-C13 | F12-E4-S10 | **NP** | **Newtown Pippen**<br>Tag AL-G Orch-111; inventory name: Newton Yellow Pippin<br>Very late; dull green-yellow; tart, aromatic winter apple<br>Cultivar profile | Apple | M-111 | FG 3 | Self-fertile | Self-fertile Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>Franklin Cider (R2-C12, 21.9 ft) - Bloom-group candidate; Harrison (R4-C12, 21.9 ft) - Bloom-group candidate<br>Inventory match: Tag AL-G Orch-111 (Grandpa's Orchard, M-111) matches the workbook. | 58 (38) |
| R4-C1 | A11-E47-S40 | **CO** | **Cox's Orange Pippin**<br>Tag AL-O Pip; inventory name: Cox Orange Pippin<br>Mid; orange-red flush; aromatic, sweet-tart; fussy<br>Cultivar profile | Apple | M-7 | FG 3 | Self-fertile | Self-fertile Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>Cortland (R3-C2, 21.9 ft) - Bloom-group candidate; Idared (R5-C2, 21.9 ft) - Bloom-group candidate | 30 (11) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R4-C2 | B11-E15-S40 | **UNK** | **Unidentified apple**<br>Tag AL-UnKn; inventory name: Unknown | Apple | Not recorded | FG ? | Not recorded | Unidentified Planted<br>Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag AL-UnKn. Owner (Oct 16, 2026): every purchased apple arrived, so a purchase with no tagged tree is presumed to have died in Boise, unrecorded. These 6 are among the 11 purchased apples with no tagged tree: Braeburn 1, Cortland 2, Dolgo 1 (long shot), Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Identify them at bloom or harvest using the possibility list.<br>Unidentified apple possibility list | 62 |
| R4-C3 | B11-E33-S40 | **HC** | **Honeycrisp (Super Red)**<br>Tag AL-Hny Crisp; inventory name: Honeycrisp, Super Red<br>Mid; red over yellow; very crisp, sweet-tart<br>Cultivar profile | Apple | M-7 | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Golden Delicious (R4-C4, 18.0 ft) - Bloom-group candidate; Hudson's Golden Gem (R5-C3, 21.9 ft) - Workbook-listed + bloom group | 50 (29) |
| R4-C4 | C11-E1-S40 | **GD** | **Golden Delicious**<br>Tag AL-Gold D; inventory name: Delicious, Golden<br>Late; golden-green; sweet, mild honey; keeps well<br>Cultivar profile | Apple | M-111 | FG 4 | Partly Self-fertile | ★ Good donor Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Honeycrisp (Super Red) (R4-C3, 18.0 ft) - Workbook-listed + bloom group; Pink Lady (Cripps Pink) (R3-C5, 21.9 ft) - Workbook-listed + bloom group | 36 (21) |
| R4-C5 | C11-E19-S40 | **UNK** | **Unidentified apple**<br>Tag AL-UnKn; inventory name: Unknown | Apple | Not recorded | FG ? | Not recorded | Unidentified Planted<br>Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag AL-UnKn. Owner (Oct 16, 2026): every purchased apple arrived, so a purchase with no tagged tree is presumed to have died in Boise, unrecorded. These 6 are among the 11 purchased apples with no tagged tree: Braeburn 1, Cortland 2, Dolgo 1 (long shot), Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Identify them at bloom or harvest using the possibility list.<br>Unidentified apple possibility list | 62 |
| R4-C6 | C11-E37-S40 | **ZT** | **Zestar**<br>Tag AL-Zestar; inventory name: Zestar<br>Earliest; rosy red blush; sweet-tart, brown sugar<br>Cultivar profile | Apple | M-111 | FG 1 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Wickson Crab (R4-C7, 18.0 ft) - Bloom-group candidate; Macintosh (R5-C7, 21.9 ft) - Bloom-group candidate<br>Inventory match: 2 of 3 purchased found. | 66 (45) |
| R4-C7 | D11-E5-S40 | **WC** | **Wickson Crab**<br>Tag AL-Wickson cr; inventory name: Crabapple, Wickson<br>Late; golf-ball red crab; very sweet, sharp; early bloom<br>Cultivar profile | Apple | B-9 | FG 2 | Not self-fertile | ★ Good donor Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Zestar (R4-C6, 18.0 ft) - Bloom-group candidate; Macintosh (R5-C7, 21.9 ft) - Bloom-group candidate | 33 (44) |
| R4-C8 | D11-E23-S40 | **UNK** | **Unidentified apple**<br>Tag AL-UnKn; inventory name: Unknown | Apple | Not recorded | FG ? | Not recorded | Unidentified Planted<br>Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag AL-UnKn. Owner (Oct 16, 2026): every purchased apple arrived, so a purchase with no tagged tree is presumed to have died in Boise, unrecorded. These 6 are among the 11 purchased apples with no tagged tree: Braeburn 1, Cortland 2, Dolgo 1 (long shot), Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Identify them at bloom or harvest using the possibility list.<br>Unidentified apple possibility list | 62 |
| R4-C9 | D11-E41-S40 | **UG** | **Tandgy green Urban Columnar - NOT IN ORCHARD**<br>Tag AL-UrbGrn | Apple | 10x3 | Midseason; FG ? | Not self-fertile | R4-C9: Tandgy green Urban Columnar (tag AL-UrbGrn, inventory row 64) not planted in the orchard. This living tree is planted outside the orchard; SiteRef D13-E0-S20. Leave this spot open; the plan is a new Haralson (apple) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock M.7, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan. | 64 (42) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R4-C10 | E11-E9-S40 | **UNK** | **Unidentified apple**<br>Tag AL-UnKn; inventory name: Unknown | Apple | Not recorded | FG ? | Not recorded | Unidentified Planted<br>Unidentified apple: bloom group and ploidy are unknown, so it is not counted as a pollen donor. Placed among identified apples of several bloom groups; identify it at bloom or harvest, then re-check its partners.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag AL-UnKn. Owner (Oct 16, 2026): every purchased apple arrived, so a purchase with no tagged tree is presumed to have died in Boise, unrecorded. These 6 are among the 11 purchased apples with no tagged tree: Braeburn 1, Cortland 2, Dolgo 1 (long shot), Empire 1, Aztec Fuji 1, Gala 1, Golden Russet 1, Honeycrisp 1, Sweet Sixteen 1, Zestar 1. Identify them at bloom or harvest using the possibility list.<br>Unidentified apple possibility list | 62 |
| R4-C11 | E11-E27-S40 | **AF** | **Auvil Early Fuji**<br>Tag AL-Fuji; inventory name: Fuji, Early Mid; red, gold-touched; sweet, low-acid Fuji sport<br>Cultivar profile | Apple | M-111 | FG 4 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Harrison (R4-C12, 18.0 ft) - Bloom-group candidate; CrimsonCrisp (R5-C12, 21.9 ft) - Bloom-group candidate<br>Inventory match: Same tag text (AL-Fuji) as the Fuji; check the variety label when planting. | 44 (3) |
| R4-C12 | E11-E45-S40 | **HR** | **Harrison**<br>Tag AL-Harrison; inventory name: Harrison<br>Late; small yellow; sweet, rich cider apple<br>Cultivar profile | Apple | M-111 or Semi Dwarf | FG 3* | Not self-fertile | Planted<br>Orange Pippin assigns Harrison to group 3, matching the existing Harrison workbook entry. The same assignment is used for Harrison Cider Apple, not a different pollen partner. Raintree instead lists group 4 (mid-late); retain this source difference and verify local bloom.<br>Auvil Early Fuji (R4-C11, 18.0 ft) - Bloom-group candidate; CrimsonCrisp (R5-C12, 21.9 ft) - Bloom-group candidate<br>Inventory match: The workbook lists one Harrison (row 26, M-111) and one Harrison Cider (row 27, semi-dwarf), the same cultivar. The tags do not say which is which, so both use the row 26 cultivar data.<br>Bloom research and sources | 48 (26, 27) |
| R4-C13 | F11-E13-S40 | **DLG** | **Dolgo Crabapple**<br>Tag AL-Dolgo; inventory name: Crabapple, Dolgo<br>Early crab; crimson, oblong; tart, white bloom<br>Cultivar profile | Apple | Not recorded | FG 2 | Not recorded | Planted<br>Orange Pippin explicitly assigns the named Dolgo cultivar to flowering group 2 and describes early blossom. This assignment assumes these are the named cultivar, not unnamed seedlings. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Harrison (R4-C12, 18.0 ft) - Workbook-listed + bloom group; Newtown Pippen (R3-C13, 21.9 ft) - Workbook-listed + bloom group<br>Inventory match: 1 of 2 purchased found.<br>Bloom research and sources | 31 (13) |
| R5-C1 | A11-E38-S20 | **CAM** | **Campfield Cider Apple**<br>Tag AL-Campfield; inventory name: Campfield Cide<br>Very late; red-striped; bittersweet Newark cider apple<br>Cultivar profile | Apple | Semi Dwarf | FG 3 | Not recorded | Planted<br>Orange Pippin explicitly assigns Campfield to flowering group 3. Raintree independently lists bloom group 3 (midseason). Both refer to flowering, not harvest season. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Idared (R5-C2, 18.0 ft) - Bloom-group candidate; Cox's Orange Pippin (R4-C1, 21.9 ft) - Workbook-listed + bloom group<br>Bloom research and sources | 28 (9) |
| R5-C2 | B11-E6-S20 | **ID** | **Idared**<br>Tag AL-Idared; inventory name: Idared<br>Late; smooth red; mild sweet-tart; keeps 6 months<br>Cultivar profile | Apple | M-111 | FG 2 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Hudson's Golden Gem (R5-C3, 18.0 ft) - Workbook-listed + bloom group; Cox's Orange Pippin (R4-C1, 21.9 ft) - Workbook-listed + bloom group | 52 (31) |
| R5-C3 | B11-E24-S20 | **HG** | **Hudson's Golden Gem**<br>Tag AL-Gold Gem; inventory name: Hudsons Golden Gem<br>Late; large russet, pear-shaped; sweet, nutty<br>Cultivar profile | Apple | M-111 | FG 3 | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Idared (R5-C2, 18.0 ft) - Bloom-group candidate; Granny Smith (R6-C2, 21.9 ft) - Workbook-listed + bloom group | 51 (30) |
| R5-C4 | B11-E42-S20 | **LB** | **Liberty**<br>Tag AL-Liberty; inventory name: Liberty Mid; bright red Mac-type; scab-resistant; sweet-tart<br>Cultivar profile | Apple | M-111 | FG 2 | Not self-fertile | Disputed Planted<br>Liberty ploidy/donor status is disputed. Excluded as a donor; given two diploid candidates.<br>Hudson's Golden Gem (R5-C3, 18.0 ft) - Bloom-group candidate; Sweet Sixteen (R5-C5, 18.0 ft) - Bloom-group candidate | 55 (35) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R5-C5 | C11-E10-S20 | **SS** | **Sweet Sixteen**<br>Tag AL- S Six; inventory name: Sweet Sixteen<br>Mid; pink-red stripes; yellow flesh, candy-sweet<br>Cultivar profile | Apple | M-7 | FG 3 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Cortland (R5-C6, 18.0 ft) - Bloom-group candidate; Golden Delicious (R4-C4, 21.9 ft) - Bloom-group candidate<br>Inventory match: 1 of 2 purchased found. | 60 (41) |
| R5-C6 | C11-E28-S20 | **CL** | **Cortland**<br>Tag AL-Cort/ AL-Cortland; inventory name: Cortland<br>Mid; red-streaked; white flesh slow to brown<br>Cultivar profile | Apple | M-111 | FG 3 | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Macintosh (R5-C7, 18.0 ft) - Bloom-group candidate; Sweet Sixteen (R5-C5, 18.0 ft) - Bloom-group candidate<br>Inventory match: 3 of 5 purchased found. | 29 (10) |
| R5-C7 | C11-E46-S20 | **MT** | **Macintosh**<br>Tag AL-Mc In; inventory name: McIntosh<br>Mid; red over green; crisp, wine-like, white flesh<br>Cultivar profile | Apple | Semi Dwarf | FG 2 | semi-self fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Cortland (R5-C6, 18.0 ft) - Bloom-group candidate; Zestar (R5-C8, 18.0 ft) - Bloom-group candidate | 57 (36) |
| R5-C8 | D11-E14-S20 | **ZT** | **Zestar**<br>Tag AL-Zestar; inventory name: Zestar Earliest; rosy red blush; sweet-tart, brown sugar<br>Cultivar profile | Apple | M-111 | FG 1 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Macintosh (R5-C7, 18.0 ft) - Bloom-group candidate; Wickson Crab (R4-C7, 21.9 ft) - Bloom-group candidate<br>Inventory match: 2 of 3 purchased found. | 66 (45) |
| R5-C9 | D11-E32-S20 | **UR** | **Tasty Red Urban Columnar - NOT IN ORCHARD**<br>Tag AL-UrbR | Apple | 10x3 (MM106) | Midseason; FG ? | Not self-fertile | R5-C9: Tasty Red Urban Columnar (tag AL-UrbR, inventory row 65) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E46-S20. Leave this spot open; the plan is a new Williams' Pride (apple) in 2027. It is ordered and has not arrived: Cummins Nursery order 039970, 2026-10-27, ships to Idaho. Rootstock B.118, grade 1, Spring 2027. Described in the planting plan. | 65 (43) |
| R5-C10 | E11-E0-S20 | **UD** | **Blushing Delight Urban Columnar - NOT IN ORCHARD**<br>Tag AL-UrbGld | Apple | 10x3 | Midseason; FG ? | Not self-fertile | R5-C10: Blushing Delight Urban Columnar (tag AL-UrbGld, inventory row 63) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E42-S20. Leave this spot open; the plan is a new Priscilla (apple) in 2027. Described in the planting plan. | 63 (6) |
| R5-C11 | E11-E18-S20 | **HR** | **Harrison**<br>Tag AL-Harrison; inventory name: Harrison<br>Late; small yellow; sweet, rich cider apple<br>Cultivar profile | Apple | M-111 or Semi Dwarf | FG 3* | Not self-fertile | Planted<br>Orange Pippin assigns Harrison to group 3, matching the existing Harrison workbook entry. The same assignment is used for Harrison Cider Apple, not a different pollen partner. Raintree instead lists group 4 (mid-late); retain this source difference and verify local bloom.<br>CrimsonCrisp (R5-C12, 18.0 ft) - Bloom-group candidate; Auvil Early Fuji (R4-C11, 21.9 ft) - Bloom-group candidate<br>Inventory match: The workbook lists one Harrison (row 26, M-111) and one Harrison Cider (row 27, semi-dwarf), the same cultivar. The tags do not say which is which, so both use the row 26 cultivar data.<br>Bloom research and sources | 48 (26, 27) |
| R5-C12 | E11-E36-S20 | **CC** | **CrimsonCrisp**<br>Tag AL-Crimson; inventory name: Crimson Crisp<br>Mid; bright crimson; very crisp, honeyed; scab-immune<br>Cultivar profile | Apple | M-111 | FG 3 | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Harrison (R5-C11, 18.0 ft) - Workbook-listed + bloom group; Auvil Early Fuji (R4-C11, 21.9 ft) - Workbook-listed + bloom group | 34 (12) |
| R5-C13 | F11-E4-S20 | **KER** | **Kerr Crab Seedling**<br>Tag AL-Kerr Crab; inventory name: Crabapple, Kerr<br>Mid; crimson-maroon crab; crisp, tart-sweet; keeps<br>Cultivar profile | Apple | Not recorded | FG 2? | Not recorded | Planted<br>BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>CrimsonCrisp (R5-C12, 18.0 ft) - Bloom-group candidate; provisional flowering group; Harrison (R4-C12, 21.9 ft) - Bloom-group candidate; provisional flowering group<br>Inventory match: 3 found, but 2 were purchased. All are seed-grown, so each may bloom differently.<br>Bloom research and sources | 32 (33) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R6-C1 | A11-E47-S0 | **KER** | **Kerr Crab Seedling**<br>Tag AL-Kerr Crab; inventory name: Crabapple, Kerr<br>Mid; crimson-maroon crab; crisp, tart-sweet; keeps<br>Cultivar profile | Apple | Not recorded | FG 2? | Not recorded | Planted<br>BEST GUESS: group 2 is an owner-approved planning estimate based on the published group for the named Kerr cultivar. The owner confirmed the Kerr trees were grown from seed (3 are at Prairie), so they are genetically different from that cultivar and may bloom at different times, including from each other. This is not a verified assignment. Record each seedling's first, full, and final bloom alongside known cultivars and revise the group if needed. Pollen-donor status remains unverified in this plan; a flowering group alone does not establish it.<br>Granny Smith (R6-C2, 18.0 ft) - Bloom-group candidate; provisional flowering group; Idared (R5-C2, 21.9 ft) - Bloom-group candidate; provisional flowering group<br>Inventory match: 3 found, but 2 were purchased. All are seed-grown, so each may bloom differently.<br>Bloom research and sources | 32 (33) |
| R6-C2 | B11-E15-S0 | **GS** | **Granny Smith**<br>Tag AL-Gran S; inventory name: Granny Smith<br>Very late; bright green; very tart, long keeper<br>Cultivar profile | Apple | Semi Dwarf | FG 3 | Self-fertile | Self-fertile ★ Good donor Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>Hudson's Golden Gem (R5-C3, 21.9 ft) - Bloom-group candidate; Idared (R5-C2, 21.9 ft) - Bloom-group candidate | 47 (25) |
| R6-C3 | B11-E33-S0 | **RS** | **M-111 apple rootstock - PLANTED ROOTSTOCK**<br>Tag AL-Rstk M111 | Apple | M-111 | No scion recorded | Not recorded | M-111 apple rootstock planted at R6-C3. No scion cultivar is recorded; not counted as a fruit tree or pollen partner until grafting and cultivar identification are recorded. | 56 |
| R7-C1 | A10-E38-S30 | **YHC** | **Yellow Huffcapp European Perry Pear**<br>Tag PR- Yw Huffcapp; inventory name: Yellow Huffcapp<br>Mid; small elliptical yellow-green; sharp perry<br>Cultivar profile | Pear | Semi Dwarf | Unconfirmed | Not recorded | Planted<br>Yellow Huffcap bloom and compatibility need confirmation; do not assume all perry pears overlap.<br>Unconfirmed — no screened partner in the available data. | 110 (108) |
| R7-C2 | B10-E6-S30 | **SA** | **Shinseiki - Asian**<br>Tag PR-Shin Asian; inventory name: Asian, Shinseiki<br>Mid; yellow skin; crisp, sweet, slightly tart<br>Cultivar profile | Pear | Semi Dwarf | Unconfirmed | Partly self-fertile | Planted<br>Shinseiki is grouped near the other Asian pears; a named compatible pair remains unconfirmed.<br>Unconfirmed — no screened partner in the available data. | 95 (103) |
| R7-C3 | B10-E24-S30 | **OG** | **Olympic Giant - Asian**<br>Tag PR-Olympic; inventory name: Olympic Giant<br>Late; huge brown russet; sweet, crisp, long keeper<br>Cultivar profile | Pear | Standard | Unconfirmed | Partly self-fertile | Planted<br>Confirm Asian-pear nursery identity and bloom. The two Giant selections are not counted as independent partners.<br>Unconfirmed — no screened partner in the available data. | 106 (99) |
| R7-C4 | B10-E42-S30 | **RB** | **Red Bartlett**<br>Tag PR-R Bart; inventory name: Bartlett, Red<br>Mid; red-skinned Bartlett sport; sweet, juicy<br>Cultivar profile | Pear | Not recorded | Early–mid | Not recorded | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Anjou (R7-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Hendre Huffcapp European Perry Pear (R8-C4, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility | 98 (102) |
| R7-C5 | C10-E10-S30 | **AU** | **Anjou**<br>Tag PR-Anjou; inventory name: Anjou Stan. Supreme<br>Late; green skin, blunt shape; mild, buttery; long keeper<br>Cultivar profile | Pear | Standard | Early–mid | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Comice (R7-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Red Bartlett (R7-C4, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility<br>Inventory match: Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased. | 92 (89) |
| R7-C6 | C10-E28-S30 | **CE** | **Comice**<br>Tag PR-Comice; inventory name: Comice<br>Late; squat, green-yellow, red blush; buttery, aromatic<br>Cultivar profile | Pear | Semi Dwarf | Mid–late | Self-fertile | Self-fertile Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>Anjou (R7-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Moonglow (R8-C6, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility | 100 (94) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R7-C7 | C10-E46-S30 | **BNT** | **Barnett European Perry Pear**<br>Tag PR-Barnet; inventory name: Barnet Euro<br>Late; small, russeted all over; light perry<br>Cultivar profile | Pear | Semi Dwarf | Unconfirmed | Not recorded | Planted<br>Confirm whether Barnett is Barnet. Barnet is very late in the perry chart; no partner is assumed.<br>Unconfirmed — no screened partner in the available data. | 96 (90) |
| R7-C8 | D10-E14-S30 | **SMC** | **SummerCrisp**<br>Tag PR-Sum Cris; inventory name: Summercrisp<br>Early; green, red blush; crisp, mild-sweet<br>Cultivar profile | Pear | Not recorded | Unconfirmed | Not recorded | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Parker (R7-C9, 18.0 ft) - Nursery-recommended Parker for Summercrisp; confirm local bloom | 109 (106) |
| R7-C9 | D10-E32-S30 | **PKR** | **Parker**<br>Tag PR-Parker; inventory name: Parker<br>Mid; yellow-bronze, red blush; hardy, tender-juicy<br>Cultivar profile | Pear | Not recorded | Unconfirmed | Not recorded | Planted<br>Parker is recommended for Summercrisp; the reverse pairing was not verified.<br>Unconfirmed — no screened partner in the available data. | 107 (100) |
| R7-C10 | E10-E0-S30 | **SMC** | **SummerCrisp**<br>Tag PR-Sum Cris; inventory name: Summercrisp<br>Early; green, red blush; crisp, mild-sweet<br>Cultivar profile | Pear | Not recorded | Unconfirmed | Not recorded | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Parker (R7-C9, 18.0 ft) - Nursery-recommended Parker for Summercrisp; confirm local bloom | 109 (106) |
| R7-C11 | E10-E18-S30 | **CE** | **Comice**<br>Tag PR-Comice; inventory name: Comice<br>Late; squat, green-yellow, red blush; buttery, aromatic<br>Cultivar profile | Pear | Semi Dwarf | Mid–late | Self-fertile | Self-fertile Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>Keiffer (R7-C12, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Anjou (R8-C10, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility | 100 (94) |
| R7-C12 | E10-E36-S30 | **KF** | **Keiffer**<br>Tag PR-Kief; inventory name: Kieffer<br>Late; yellow, crimson blush; crisp, coarse; canning pear<br>Cultivar profile | Pear | Not recorded | Late | Not recorded | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Comice (R7-C11, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Butt European Perry Pear (R7-C13, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | 104 (97) |
| R7-C13 | F10-E4-S30 | **BUT** | **Butt European Perry Pear**<br>Tag PR-Butt Euro; inventory name: Butt Euro Perry<br>Late; small lemon-yellow; astringent perry pear<br>Cultivar profile | Pear | Semi Dwarf | Late (perry chart) | Not recorded | Planted<br>Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility.<br>Keiffer (R7-C12, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Comice (R8-C12, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility | 99 (93) |
| R8-C1 | A10-E47-S10 | **PE** | **Pineapple - DEAD**<br>Tag PR-Pineap | Pear | Semi Dwarf | Early | Self-fertile | R8-C1: Pineapple (tag PR-Pineap, inventory row 108) reported dead October 11, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Moonglow, stand-in for Harrow Delight (pear) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock OHxF 97, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan. | 108 (101) |
| R8-C2 | B10-E15-S10 | **HA** | **Hosui - Asian**<br>Tag PR-Asian H; inventory name: Asian Hosui<br>Mid; round golden-russet; sweet-tart, very juicy<br>Cultivar profile | Pear | Standard | Early | Not self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Starking Hardy Giant - Asian (R8-C3, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | 93 (96) |
| R8-C3 | B10-E33-S10 | **HG** | **Starking Hardy Giant - Asian**<br>Tag PR-Asian; inventory name: Asian, Hardy Giant<br>Late; huge round brown russet; crisp, long keeper<br>Cultivar profile | Pear | Semi Dwarf | Early | Not self-fertile | Planted<br>Confirm Asian-pear nursery identity and bloom. The two Giant selections are not counted as independent partners.<br>Hosui - Asian (R8-C2, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | 94 (105) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R8-C4 | C10-E1-S10 | **HHC** | **Hendre Huffcapp European Perry Pear**<br>Tag PR-Hendre; inventory name: Hendre Huffeap<br>Late; small elliptical; light low-tannin perry<br>Cultivar profile | Pear | Semi Dwarf | Early (perry chart) | Not recorded | Planted<br>Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility.<br>Bartlett (R8-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Anjou (R7-C5, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility<br>Inventory match: 2 found, but 1 was purchased. Thorn (row 107) was not found; one of these may be Thorn. | 103 (95) |
| R8-C5 | C10-E19-S10 | **BT** | **Bartlett**<br>Tag PR-Bart; inventory name: Bartlett<br>Mid; green turning yellow; sweet, juicy, aromatic<br>Cultivar profile | Pear | Standard | Early–mid | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Hendre Huffcapp European Perry Pear (R8-C4, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Moonglow (R8-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | 97 (91, 92) |
| R8-C6 | C10-E37-S10 | **MW** | **Moonglow**<br>Tag PR-Moon; inventory name: Moonglow<br>Mid; yellow, pink blush; soft, juicy, mild<br>Cultivar profile | Pear | Standard | Early–mid | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Anjou (R8-C7, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Bartlett (R8-C5, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | 105 (98) |
| R8-C7 | D10-E5-S10 | **AU** | **Anjou**<br>Tag PR-D'an; inventory name: D'Anjou<br>Late; green skin, blunt shape; mild, buttery; long keeper<br>Cultivar profile | Pear | Standard | Early–mid | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Hendre Huffcapp European Perry Pear (R8-C8, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Moonglow (R8-C6, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility<br>Inventory match: Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased. | 101 (89) |
| R8-C8 | D10-E23-S10 | **HHC** | **Hendre Huffcapp European Perry Pear**<br>Tag PR-Hendre; inventory name: Hendre Huffeap<br>Late; small elliptical; light low-tannin perry<br>Cultivar profile | Pear | Semi Dwarf | Early (perry chart) | Not recorded | Planted<br>Perry bloom window is from UK observations; confirm cultivar identity, local bloom, and genetic compatibility.<br>Anjou (R8-C7, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Starking Delicious (R8-C9, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility<br>Inventory match: 2 found, but 1 was purchased. Thorn (row 107) was not found; one of these may be Thorn. | 103 (95) |
| R8-C9 | D10-E41-S10 | **SD** | **Starking Delicious**<br>Tag PR-Delic; inventory name: Delicious<br>Mid; large yellow-green; sweet, juicy, easy to peel<br>Cultivar profile | Pear | Standard | Early–mid | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Anjou (R8-C10, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Hendre Huffcapp European Perry Pear (R8-C8, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | 102 (104) |
| R8-C10 | E10-E9-S10 | **AU** | **Anjou**<br>Tag PR-D'an; inventory name: D'Anjou<br>Late; green skin, blunt shape; mild, buttery; long keeper<br>Cultivar profile | Pear | Standard | Early–mid | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Starking Delicious (R8-C9, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Bartlett (R8-C11, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility<br>Inventory match: Inventory rows 92 (Anjou) and 101 (D'Anjou) together hold 3 Anjou, but 2 were purchased. | 101 (89) |
| R8-C11 | E10-E27-S10 | **BT** | **Bartlett**<br>Tag PR-Bart; inventory name: Bartlett<br>Mid; green turning yellow; sweet, juicy, aromatic<br>Cultivar profile | Pear | Standard | Early–mid | Partly self-fertile | Planted<br>Confirm overlapping bloom locally before relying on these candidates.<br>Comice (R8-C12, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Anjou (R8-C10, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility | 97 (91, 92) |
| R8-C12 | E10-E45-S10 | **CE** | **Comice**<br>Tag PR-Comice; inventory name: Comice<br>Late; squat, green-yellow, red blush; buttery, aromatic<br>Cultivar profile | Pear | Semi Dwarf | Mid–late | Self-fertile | Self-fertile Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>Bartlett (R8-C11, 18.0 ft) - Pear bloom-chart candidate; confirm compatibility; Keiffer (R7-C12, 21.9 ft) - Pear bloom-chart candidate; confirm compatibility | 100 (94) |
| R9-C1 | A9-E38-S40 | **BB** | **Bing**<br>Tag CR-Bing; inventory name: Bing<br>Mid; mahogany to near-black heart; very sweet, firm<br>Cultivar profile | Cherry | Semi Dwarf | Cherry period 3 | Partly self-fertile | Planted Pseudomonas +<br>Treat Bing as cross-pollination dependent. Stella is conservatively excluded as its donor.<br>Lapins (R10-C2, 33.6 ft) - Cherry compatibility + bloom chart; confirm local overlap<br>Pseudomonas positive, untreated; 1 of the 2 trees with this tag. | 72 (53, 54) |
| R9-C2 | B9-E6-S40 | **RS** | **Royalton Sweet - DEAD**<br>Tag CR-Royalton | Cherry | Semi Dwarf | Cherry period 3 | Partly self-fertile | R9-C2: Royalton Sweet (tag CR-Royalton, inventory row 80) reported dead October 17, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Black Pearl (cherry) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order XFEEV5ANY, 2026-10-28, ships to Idaho. Rootstock Mazzard, grade Hand Select, Spring 2027. Described in the planting plan. | 80 (63) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R9-C3 | B9-E24-S40 | **BB** | **Bing**<br>Tag CR-Bing; inventory name: Bing<br>Mid; mahogany to near-black heart; very sweet, firm<br>Cultivar profile | Cherry | Semi Dwarf | Cherry period 3 | Partly self-fertile | Planted Pseudomonas +<br>Treat Bing as cross-pollination dependent. Stella is conservatively excluded as its donor.<br>Lapins (R10-C2, 21.9 ft) - Cherry compatibility + bloom chart; confirm local overlap<br>Pseudomonas positive, untreated; 1 of the 2 trees with this tag. | 72 (53, 54) |
| R9-C4 | B9-E42-S40 | **ST** | **Stella**<br>Tag CR-Stella; inventory name: Sweet Stella<br>Mid; self-fertile; large dark heart-shaped, firm<br>Cultivar profile | Cherry | Semi Dwarf | Cherry period 3 | self-fertile | Self-fertile Planted Pseudomonas +<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>Bing (R9-C3, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap; Lapins (R10-C2, 33.6 ft) - Cherry compatibility + bloom chart; confirm local overlap<br>Pseudomonas positive, untreated. | 82 (65) |
| R9-C5 | C9-E10-S40 | **NKI** | **Nanking (bush)**<br>Tag CR-Ian Nan; inventory name: Nanking, Ian<br>Mid; small tart-sweet cherries; very hardy bush<br>Cultivar profile | Cherry | Not recorded | Unconfirmed | Not recorded | Planted<br>Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Nanking 'Ian' (2) and 'Jules' (1) together match the 3 purchased Nanking bushes. | 78 (61) |
| R9-C6 | C9-E28-S40 | **SWT** | **Unidentified sweet cherry**<br>Tag CR-Sweet; inventory name: Sweet | Cherry | Not recorded | Unconfirmed | Not recorded | Unidentified Planted Pseudomonas +<br>Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag CR-Sweet. Purchased sweet cherries not otherwise found at Prairie: Black Tartarian 2, Regina 1, Sam 1.<br>Unidentified sweet cherry possibility list<br>Pseudomonas positive, untreated; 1 of the 3 trees with this tag. | 81 |
| R9-C7 | C9-E46-S40 | **HG** | **Hardy Giant Antque Sweet**<br>Tag CR-Hardy Antq; inventory name: Hardy Giant Antique<br>Mid; mahogany skin, dark red flesh; heavy clusters<br>Cultivar profile | Cherry | Semi Dwarf | Unconfirmed | Partly self-fertile | Planted Pseudomonas +<br>Sweet-cherry donor compatibility was not verified; do not rely on this tree to supply other varieties.<br>Unconfirmed — no screened partner in the available data.<br>Pseudomonas positive, untreated. | 75 (58) |
| R9-C8 | D9-E14-S40 | **NKI** | **Nanking (bush)**<br>Tag CR-Ian Nan; inventory name: Nanking, Ian<br>Mid; small tart-sweet cherries; very hardy bush<br>Cultivar profile | Cherry | Not recorded | Unconfirmed | Not recorded | Planted<br>Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Nanking 'Ian' (2) and 'Jules' (1) together match the 3 purchased Nanking bushes. | 78 (61) |
| R10-C1 | A9-E47-S20 | **BT** | **Black Tartarian**<br>Tag CR-Black T; inventory name: Black Tartarian<br>Early; purple-black, medium; sweet, rich; pollinizer<br>Cultivar profile | Cherry | Standard | Cherry period 1 | Partly self-fertile | Planted Pseudomonas +<br>Black Tartarian blooms very early; Lapins is its only identified donor in an adjacent bloom period. Two of the four purchased Black Tartarians were not found and may be among the unidentified sweet cherries.<br>Lapins (R10-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap<br>Inventory match: 2 of 4 purchased found.<br>Pseudomonas positive, untreated; 1 of the 2 trees with this tag. | 73 (55, 56) |
| R10-C2 | B9-E15-S20 | **LP** | **Lapins**<br>Tag CR-Lapins; inventory name: Lapins Sweet<br>Late; self-fertile; large mahogany, Bing-shaped<br>Cultivar profile | Cherry | Not recorded | Cherry period 2 | Partly self-fertile | Self-fertile Planted<br>Lapins is self-fertile in the reference chart and is the only identified sweet cherry within one bloom period of both Black Tartarian and Bing, so the sweet cherries are grouped around it.<br>Black Tartarian (R10-C1, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap; Bing (R9-C3, 21.9 ft) - Cherry compatibility + bloom chart; confirm local overlap | 76 (59) |
| R10-C3 | B9-E33-S20 | **BT** | **Black Tartarian**<br>Tag CR-Black T; inventory name: Black Tartarian<br>Early; purple-black, medium; sweet, rich; pollinizer<br>Cultivar profile | Cherry | Standard | Cherry period 1 | Partly self-fertile | Planted Pseudomonas +<br>Black Tartarian blooms very early; Lapins is its only identified donor in an adjacent bloom period. Two of the four purchased Black Tartarians were not found and may be among the unidentified sweet cherries.<br>Lapins (R10-C2, 18.0 ft) - Cherry compatibility + bloom chart; confirm local overlap<br>Inventory match: 2 of 4 purchased found.<br>Pseudomonas positive, untreated; 1 of the 2 trees with this tag. | 73 (55, 56) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R10-C4 | C9-E1-S20 | **SWT** | **Unidentified sweet cherry**<br>Tag CR-Sweet; inventory name: Sweet | Cherry | Not recorded | Unconfirmed | Not recorded | Unidentified · Planted · Pseudomonas +<br>Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag CR-Sweet. Purchased sweet cherries not otherwise found at Prairie: Black Tartarian 2, Regina 1, Sam 1.<br>Unidentified sweet cherry possibility list<br>Pseudomonas positive, untreated; 1 of the 3 trees with this tag. | 81 |
| R10-C5 | C9-E19-S20 | **MNT** | **Montmorency**<br>Tag CR-Mont; inventory name: Montmorency<br>Late; bright red tart pie cherry; self-fertile<br>Cultivar profile | Cherry | Not recorded | Unconfirmed | Not recorded | Self-fertile · Planted<br>Montmorency is a self-fertile tart cherry; not used in the sweet-cherry donor network.<br>No extra partner required under the current self-fertility guidance. | 77 (60) |
| R10-C6 | C9-E37-S20 | **NKJ** | **Nanking (bush)**<br>Tag CR-Juls Nan; inventory name: Nanking, Jules<br>Mid; small tart-sweet cherries; very hardy bush<br>Cultivar profile | Cherry | Not recorded | Unconfirmed | Not recorded | Planted<br>Nanking bush cherry: needs a genetically different Nanking with overlapping bloom and is not a sweet-cherry pollen donor. The Ian and Jules plants are kept together in case they are different seedling lines.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Nanking 'Ian' (2) and 'Jules' (1) together match the 3 purchased Nanking bushes. | 79 (61) |
| R10-C7 | D9-E5-S20 | **SWT** | **Unidentified sweet cherry**<br>Tag CR-Sweet; inventory name: Sweet | Cherry | Not recorded | Unconfirmed | Not recorded | Unidentified · Planted · Pseudomonas +<br>Unidentified sweet cherry, possibly one of the purchased Black Tartarian, Regina, or Sam trees not otherwise found. Its compatibility group is unknown, so it is not counted as a donor until identified.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag CR-Sweet. Purchased sweet cherries not otherwise found at Prairie: Black Tartarian 2, Regina 1, Sam 1.<br>Unidentified sweet cherry possibility list<br>Pseudomonas positive, untreated; 1 of the 3 trees with this tag. | 81 |
| R10-C8 | D9-E23-S20 | **MNT** | **Montmorency**<br>Tag CR-Mont; inventory name: Montmorency<br>Late; bright red tart pie cherry; self-fertile<br>Cultivar profile | Cherry | Not recorded | Unconfirmed | Not recorded | Self-fertile · Planted<br>Montmorency is a self-fertile tart cherry; not used in the sweet-cherry donor network.<br>No extra partner required under the current self-fertility guidance. | 77 (60) |
| R11-C1 | A9-E38-S0 | **MP** | **Moorpark**<br>Tag AT-moorpark; inventory name: Moore Park<br>Mid; large orange, red-specked; rich flavor; fresh, dried<br>Cultivar profile | Apricot | Not recorded | Unconfirmed | Self-fertile | Self-fertile · Planted · Pseudomonas +<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Inventory match: 2 found, but 1 Moorpark was purchased. The extra Moorpark tag is still unexplained. The former AT-Unk tree (row 71) was identified as Patterson on 2026-10-25, so it is not one of these.<br>Pseudomonas positive, untreated; 1 of the 2 trees with this tag. | 70 (49) |
| R11-C2 | B9-E6-S0 | **PN** | **Patterson**<br>Tag AT-Patterson; inventory name: Patterson<br>Early; firm, red-streaked yellow; self-fertile; canning<br>Cultivar profile | Apricot | Standard | Unconfirmed | Self-fertile | Self-fertile · Planted · Pseudomonas +<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Inventory match: Owner identified the AT-Unk apricot as Patterson on 2026-10-25. Planted at R11-C2. Metal tag is now AT-Patterson.<br>Pseudomonas positive, untreated. | 71 (50) |
| R11-C3 | B9-E24-S0 | **GT** | **Giant Tilton**<br>Tag AT-Tilton; inventory name: Giant Tilton<br>Mid; large golden-orange; fresh, canning, drying<br>Cultivar profile | Apricot | Standard | Unconfirmed | Self-fertile | Self-fertile · Planted · Pseudomonas +<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Pseudomonas positive, untreated. | 68 (47) |
| R11-C6 | C9-E28-S0 | **IPL** | **Italian Plum**<br>Tag PM- It Lov; inventory name: Italian Lovell<br>Mid; dark blue prune plum; sweet, freestone, dries well<br>Cultivar profile | Plum | Not recorded | Unconfirmed | Not recorded | Self-fertile · Planted<br>Tag "Italian Lovell": Italian plum, probably on Lovell rootstock. The named Italian plum is self-fertile or partly self-fertile in references; if one of these is a purchased Italian seedling, its fertility is unknown.<br>No extra partner required under the current self-fertility guidance.<br>Inventory match: 3 of 4 found (2 named Italian plums, row 109; 2 Italian seedlings, row 110). Lovell is a peach rootstock, which suggests grafted trees; confirm whether any are seedlings. | 111 (109, 110) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R11-C8 | D9-E14-S0 | **CR** | **Contender**<br>Tag PC-Contender; inventory name: Contender<br>Mid; red-orange skin; very hardy; non-browning flesh<br>Cultivar profile | Peach | Standard | Unconfirmed | Self-fertile | Self-fertile Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Inventory match: 1 of 2 purchased found. | 85 (79) |
| R11-C9 | D9-E32-S0 | **RH** | **Redhaven**<br>Tag PC-Rd Haven; inventory name: Redhaven<br>Early; red-blushed, little fuzz; sweet yellow flesh<br>Cultivar profile | Peach | Semi Dwarf | Unconfirmed | Self-fertile | Self-fertile Planted Pseudomonas +<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Pseudomonas positive, untreated. | 89 (84) |
| R11-C10 | E9-E0-S0 | **JH** | **JH Hale - DEAD**<br>Tag PC-Hale | Peach | Semi Dwarf | Unconfirmed | Self-fertile | R11-C10: JH Hale (tag PC-Hale, inventory row 88) reported dead October 17, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Intrepid (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Seedling, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan. | 88 (83) |
| R11-C11 | E9-E18-S0 | **FT** | **Frost**<br>Tag PC-Frost; inventory name: Frost<br>Mid; leaf curl resistant; yellow-green skin, pink bloom<br>Cultivar profile | Peach | Semi Dwarf | Unconfirmed | Self-fertile | Self-fertile Planted Pseudomonas +<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Pseudomonas positive, untreated. | 87 (81) |
| R11-C12 | E9-E36-S0 | **CH** | **Canadian Harmony - DEAD**<br>Tag PC-Can Harm | Peach | Semi Dwarf | Unconfirmed | Self-fertile | R11-C12: Canadian Harmony (tag PC-Can Harm, inventory row 84) reported dead October 18, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Contender (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Lovell, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan. | 84 (78) |
| R11-C13 | F9-E4-S0 | **RC** | **Reliance**<br>Tag PC-Reliance; inventory name: Reliance<br>Early; very hardy, late bloom; soft, sweet, juicy<br>Cultivar profile | Peach | Standard | Unconfirmed | Self-fertile | Self-fertile Planted Pseudomonas +<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Inventory match: 2 of 3 purchased found.<br>Pseudomonas positive, untreated. | 90 (85, 86, 87) |
| R12-C1 | A8-E47-S30 | **HT** | **Harcot**<br>Tag AT-Harcot; inventory name: Harcot<br>Mid; orange, red blush; late bloom, canker resistant<br>Cultivar profile | Apricot | Semi Dwarf | Unconfirmed | Self-fertile | Self-fertile Planted Pseudomonas +<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Pseudomonas positive, untreated. | 69 (48) |
| R12-C2 | B8-E15-S30 | **MP** | **Moorpark**<br>Tag AT-moorpark; inventory name: Moore Park<br>Mid; large orange, red-specked; rich flavor; fresh, dried<br>Cultivar profile | Apricot | Not recorded | Unconfirmed | Self-fertile | Self-fertile Planted Pseudomonas +<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Inventory match: 2 found, but 1 Moorpark was purchased. The extra Moorpark tag is still unexplained. The former AT-Unk tree (row 71) was identified as Patterson on 2026-10-25, so it is not one of these.<br>Pseudomonas positive, untreated; 1 of the 2 trees with this tag. | 70 (49) |
| R12-C5 | C8-E19-S30 | **IPL** | **Italian Plum**<br>Tag PM- It Lov; inventory name: Italian Lovell<br>Mid; dark blue prune plum; sweet, freestone, dries well<br>Cultivar profile | Plum | Not recorded | Unconfirmed | Not recorded | Self-fertile Planted<br>Tag "Italian Lovell": Italian plum, probably on Lovell rootstock. The named Italian plum is self-fertile or partly self-fertile in references; if one of these is a purchased Italian seedling, its fertility is unknown.<br>No extra partner required under the current self-fertility guidance.<br>Inventory match: 3 of 4 found (2 named Italian plums, row 109; 2 Italian seedlings, row 110). Lovell is a peach rootstock, which suggests grafted trees; confirm whether any are seedlings. | 111 (109, 110) |
| R12-C6 | C8-E37-S30 | **IPL** | **Italian Plum**<br>Tag PM- It Lov; inventory name: Italian Lovell<br>Mid; dark blue prune plum; sweet, freestone, dries well<br>Cultivar profile | Plum | Not recorded | Unconfirmed | Not recorded | Self-fertile Planted<br>Tag "Italian Lovell": Italian plum, probably on Lovell rootstock. The named Italian plum is self-fertile or partly self-fertile in references; if one of these is a purchased Italian seedling, its fertility is unknown.<br>No extra partner required under the current self-fertility guidance.<br>Inventory match: 3 of 4 found (2 named Italian plums, row 109; 2 Italian seedlings, row 110). Lovell is a peach rootstock, which suggests grafted trees; confirm whether any are seedlings. | 111 (109, 110) |

| Position | Spot SiteRef | Code | Variety and tag | Fruit type | Rootstock | Bloom / FG | Fertility (workbook) | Planning notes / nearest candidates | Inventory row (Fruit row) |
|---|---|---|---|---|---|---|---|---|---|
| R12-C9 | D8-E41-S30 | **RC** | **Reliance**<br>Tag PC-Reliance; inventory name: Reliance<br>Early; very hardy, late bloom; soft, sweet, juicy<br>Cultivar profile | Peach | Standard | Unconfirmed | Self-fertile | Self-fertile · Planted · Pseudomonas +<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Inventory match: 2 of 3 purchased found.<br>Pseudomonas positive, untreated. | 90 (85, 86, 87) |
| R12-C10 | E8-E9-S30 | **UNK** | **Unidentified peach**<br>Tag PC-Unk; inventory name: Unknown | Peach | Not recorded | Unconfirmed | Not recorded | Unidentified · Planted · Pseudomonas +<br>Unidentified peach, possibly Contender, Elberta, Reliance, or Veteran. Not counted as a pollen donor until identified.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag PC-Unk. Purchased peaches not otherwise found at Prairie: Contender 1, Elberta 1, Reliance 1, Veteran 1.<br>Unidentified peach possibility list<br>Pseudomonas positive, untreated. | 91 |
| R12-C11 | E8-E27-S30 | **BS** | **BlazingStar - DEAD**<br>Tag PC-Blaz Star | Peach | Not recorded | Unconfirmed | Self-fertile | R12-C11: BlazingStar (tag PC-Blaz Star, inventory row 83) reported dead October 18, 2026 (it died in its pot before planting). Leave this spot open; the plan is a new Redhaven, stand-in for Harrow Diamond (peach) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock Seedling, grade Field Ready, Spring 2027. Payment is still due. Described in the planting plan. | 83 (77) |
| R12-C12 | E8-E45-S30 | **UNK** | **Unidentified peach**<br>Tag PC-Unk; inventory name: Unknown | Peach | Not recorded | Unconfirmed | Not recorded | Unidentified · Planted · Pseudomonas +<br>Unidentified peach, possibly Contender, Elberta, Reliance, or Veteran. Not counted as a pollen donor until identified.<br>Unconfirmed — no screened partner in the available data.<br>Inventory match: Tag PC-Unk. Purchased peaches not otherwise found at Prairie: Contender 1, Elberta 1, Reliance 1, Veteran 1.<br>Unidentified peach possibility list<br>Pseudomonas positive, untreated. | 91 |
| R12-C13 | F8-E13-S30 | **EA** | **Elberta**<br>Tag PC-Elberta; inventory name: Elberta<br>Late; large golden-yellow; sweet, firm freestone<br>Cultivar profile | Peach | Standard | Unconfirmed | Self-fertile | Self-fertile · Planted<br>Self-fertility is useful insurance; local bloom and insect activity still affect cropping.<br>No extra partner required under the current self-fertility guidance.<br>Inventory match: 1 of 2 purchased found. | 86 (80) |

## Not in the Main Orchard

Nut trees are kept out of the main orchard. Rootstock quantities below are the remaining stock, excluding the M-111 already planted at R6-C3. The three Urban apples are living trees planned for planting outside the orchard; their new grid locations have not been supplied. The Boise trees were not moved to Prairie. Windbreak, east-hedge, and berry guild plants are covered in the windbreak section and the site plan notes.
A SiteRef of PENDING means no individual location has been supplied; it does not assign one coordinate to an inventory quantity group.

| Inventory row | Item | Qty | SiteRef | Status and notes |
|---|---|---|---|---|
| 38 | **Apple — EMLA7 Apple Rootstock** · Tag AL-Rtsk-EMLA7 | 2 | PENDING | Rootstock: graft first, then plant in reserved row R1. Ungrafted EMLA 7 rootstock, for grafting. |
| 56 | **Apple — M 111 Rootstock** · Tag AL-Rstk M111 | 11 | PENDING | Rootstock: graft first, then plant in reserved row R1. Ungrafted M-111 rootstock, for grafting. 1 of the original 12 already planted at R6-C3; the quantity here is the remaining stock. |
| 61 | **Apple — Ukn Rootstock** · Tag AL-Rtsk Unk | 1 | PENDING | Rootstock: graft first, then plant in reserved row R1. Rootstock of unknown type, for grafting. |
| 67 | **Apricot — Chinese (Mormon)** · Tag AT-Chinese | 1 | PENDING | Still in Boise (girdled, mulched); not planned for Prairie. |
| 74 | **Cherry — Craigs Crimson** · Tag CR-Crimson | 1 | PENDING | Still in Boise (girdled, mulched); not planned for Prairie. |
| 112 | **Hican** · Tag hican | 10 | PENDING | Nut tree: kept out of the main orchard. |
| 113 | **Pecan** | 4 | PENDING | Nut tree: kept out of the main orchard. 4 of 10 purchased found. |
| 114 | **Walnut — English** · Tag WL-Enligsh | 4 | PENDING | Nut tree: kept out of the main orchard. Not on the Fruit sheet. The inventory notes 3 are already planted. |
| 115 | **Walnut — Heartnut Japanese** · Tag WL-Heart J | 2 | PENDING | Nut tree: kept out of the main orchard. |
| 64 | **Tandgy green Urban Columnar** · Tag AL-UrbGrn | 1 | D13-E0-S20 | R4-C9: Tandgy green Urban Columnar (tag AL-UrbGrn, inventory row 64) not planted in the orchard. This living tree is planted outside the orchard; SiteRef D13-E0-S20. Leave this spot open; the plan is a new Haralson (apple) in 2027. It is ordered and has not arrived: Roots to Fruits Nursery order RTF12645, 2026-10-27, ships to Idaho. Rootstock M.7, grade Hand Select, Spring 2027. Payment is still due. Described in the planting plan. |
| 65 | **Tasty Red Urban Columnar** · Tag AL-UrbR | 1 | C13-E46-S20 | R5-C9: Tasty Red Urban Columnar (tag AL-UrbR, inventory row 65) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E46-S20. Leave this spot open; the plan is a new Williams' Pride (apple) in 2027. It is ordered and has not arrived: Cummins Nursery order 039970, 2026-10-27, ships to Idaho. Rootstock B.118, grade 1, Spring 2027. Described in the planting plan. |

| Inventory row | Item | Qty | SiteRef | Status and notes |
|---|---|---|---|---|
| 63 | **Blushing Delight Urban Columnar** · Tag AL-UrbGld | 1 | `C13-E42-S20` | R5-C10: Blushing Delight Urban Columnar (tag AL-UrbGld, inventory row 63) not planted in the orchard. This living tree is planted outside the orchard; SiteRef C13-E42-S20. Leave this spot open; the plan is a new Priscilla (apple) in 2027. Described in the planting plan. |

M-111 apple rootstock planted at R6-C3. No scion cultivar is recorded; not counted as a fruit tree or pollen partner until grafting and cultivar identification are recorded.

3 living Urban apples are planted outside the orchard, 20 ft south of R1. Their former orchard slots stay open for new trees in 2027 and are not used for spare rootstocks now.

Penn State notes that black walnut and butternut are the best-known sources of juglone, a compound that injures many plants, and that English walnut and hickories (the group that includes pecans and hicans) also produce it to a lesser degree. Heartnut is a walnut closely related to butternut. Site these nut trees away from the orchard rather than beside it.

Once grafted, 13 of the 14 remaining rootstocks go in reserved row R1; the other 1 is a spare for a graft that fails or a later replacement. R13–R15 are held open for more trees you buy or graft later.

The 2027 planting plan names a cultivar for every open spot in R1 to R12: 49 in all (the 8 spots held for replacements, 28 other empty positions, and the 13 reserved spots in R1, which are grafted). They show as green cells, and the 32 already ordered show in blue, marked ord. None is counted as a tree or pollen partner until it is in the ground. R13–R15 stay open for later.

The Chinese apricot and Craig's Crimson cherry are still in Boise (girdled and mulched) and are not part of the Prairie plan.

## Planning Corrections and Sources

Canonical JSON in data/ supplies the current records, with the original inventory and purchase workbooks retained for provenance. Owner-reported losses and outside-orchard plans leave their original slots open and are excluded from orchard counts and pollen partners. The M-111 planted at R6-C3 is allocated from inventory row 56; 11 of that line's original 12 rootstocks remain available. It is not a fruit tree or pollen partner without a recorded scion cultivar. Every inventory line is matched to its purchase record(s) in the reconciliation table. When one inventory line matches several records of the same cultivar, the first record's cultivar data is used and the rootstocks lists the possibilities. Plot dimensions, windbreaks, hedges, juniper distance, planting updates, and white pine losses are retained in the JSON records. Berry bushes are matched to the original Berry sheet.

Invalid cross-species spreadsheet entries are not used: Bartlett/Moonglow/Anjou are not cherry pollenizers, and Jefferson/Theta are not pear pollenizers. The size note in the Asian Hardy Giant pollenizer cell is not a pollination relationship.

Planning corrections are separate from the source fields: Liberty donor status remains disputed; Lapins and Montmorency are treated as self-fertile; Bing is not treated as reliably self-fertile; J.H. Hale needed another peach despite its workbook label (it has since died). Original fertility values remain visible in the inventory.

Apple flowering-group research is dated September 24, 2026 and linked in its own table. Workbook FG values drive the grid colors, bloom labels, and candidate screening. Seed-grown Kerr uses group 2 only as an owner-approved best guess (FG 2?). Qualitative Urban Apple bloom seasons remain unnumbered.

Pseudomonas (bacterial canker) results come from the inventory. Positive, untreated trees are flagged but not relocated; the layout is driven by pollination.

Before planting, confirm cultivar identities, bloom overlap, local climate suitability, rootstock and mature size, and insect activity. Windbreak sizes and spacings are planning values from the cited guides; adjust them to the plants you have.


1. Flowering groups and overlapping bloom — Orange Pippin
2. Fruit-tree pollination, cultivar identity, and cherry/plum cautions — Cummins Nursery
3. Apple cross-compatibility and pollenizers — Washington State University
4. Liberty: disputed ploidy and pollenizing ability — Orange Pippin
5. Pear bloom and pollination chart — Stark Bro's
6. Perry-pear flowering observations — National Perry Pear Centre
7. Summercrisp: Parker named as a pollenizer — Charlie the Tree Guy
8. Sweet-cherry compatibility and bloom chart — Oregon State University
9. J.H. Hale: almost any other peach except Elberta; Veteran named — UC Davis Foundation Plant Services
10. J.H. Hale pollinizers and the Elberta exception — Ask Extension
11. Windbreak row spacing, in-row spacing, end extension, and protection distance — University of Nebraska–Lincoln EC1767
12. Hybrid hazelnut growth habit, spacing, hand harvest, and coppicing — Upper Midwest Hazelnut Development Initiative
13. Hazelnut seedlings ripening late August into October — UMHDI harvest timing bulletin
14. Hybrid hazelnuts in windbreaks: taller tree rows at least 15 ft away — USDA National Agroforestry Center
15. Theta as a Jefferson hazelnut pollenizer — Oregon State University
16. Elderberry spacing and hardwood cuttings — West Virginia University Extension
17. Currant and gooseberry spacing, sun, self-fertility, and jostaberry rust resistance — University of Minnesota Extension
18. White pine blister rust: five-needle pines and currant/gooseberry hosts — WSU HortSense
19. Honeyberries: partial shade and two varieties for pollination — University of Alaska Fairbanks Cooperative Extension
20. Comfrey: deep roots make it difficult to eradicate — WSU HortSense
21. Red mulberry size, male/female trees, and pruning time — Missouri Botanical Garden
22. Red mulberry: a tree, root sprouting, and crosses with white mulberry — USDA Forest Service (FEIS)
23. Red vs. white mulberry: open-grown form and western windbreaks — Purdue Extension FNR-237
24. Harvesting mulberries by shaking branches over a tarp — Ask Extension
25. Mulberry pruning: cuts on limbs 2 in or thicker heal poorly — Ask Extension
26. Cedar-apple rust and keeping junipers away from apples — University of Minnesota Extension
27. Juglone and nut trees — Penn State Extension


Prairie Plants Orchard — Inventory-based planting plan, October 4, 2026; planting status updated October 18, 2026 — 237 × 600 ft orchard plot plus 35 ft west windbreak, C1 berry guilds, and east elderberry hedge — 13 across — 18 × 20 ft triangular offset — Planning schematic; verify site suitability, identities, and bloom before planting.